

A Mini Project Report On
“DIGITAL DAIRY MANAGEMENT SYSTEM”

Submitted to the
Department of Computer Applications
In partial fulfillment of the

BACHELOR OF COMPUTER APPLICATION

Under the guidance of
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BONAFIDE CERTIFICATE

Certified that the Mini-Project Work entitled

“DIGITAL DAIRY MANAGEMENT SYSTEM”

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Degree From

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2023-2026

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PROJECT GUIDE

Submitted for the Viva-Voce Examination held on.....

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This is to certify that the mini-project entitled **“DIGITAL DAIRY MANAGEMENT SYSTEM”** has been successfully carried out by **Joyal Royce Abraham (Reg No:230021083059)** in partial fulfilment of the Course Department of Computer Applications.

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DECLARATION

We hereby declare that the mini-project work entitled "DIGITAL DAIRY MANAGEMENT SYSTEM" is an authenticated work carried out by us at **Mar Athanasios College for Advanced Studies Thiruvalla (MACFAST), Thiruvalla** under the guidance of **Ms. AKSHARA SAMBASIVAN** for the partial fulfilment of the course DEPARTMENT OF COMPUTER APPLICATIONS. This work has not been submitted for similar purpose anywhere else except to MAR ATHANASIOS COLLEGE FOR ADVANCED STUDIES TIRUVALLA (MACFAST), THIRUVALLA. I understand that detection of any such copying is liable to be punished in any way the school deems fit.

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JOYAL ROYCE ABRAHAM

Signature of Student

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ABSTRACT

The Digital Dairy Management System (DDMS) is a comprehensive web-based application developed to streamline and digitalize the operations of dairy farms and cooperatives. The system automates key processes such as farmer registration, milk collection, product inventory, sales management, delivery tracking, and reporting, providing a unified platform for administrators, farmers, and customers. Developed using PHP, MySQL, HTML, CSS, and Bootstrap, the system ensures efficient data management and a user-friendly interface for real-time monitoring of dairy activities. It enables administrators to manage farmers, track milk collection, maintain inventory, process customer orders, and generate analytical reports with ease. Farmers can view their daily milk supply records, track payments, and analyse performance, while customers can browse products, place online orders, and track deliveries conveniently. The main objective of DDMS is to reduce manual workload, eliminate data redundancy, and enhance transparency in dairy operations through digital automation. The system supports secure authentication, role-based access control, and data validation to ensure accuracy and reliability. It provides valuable insights into sales trends and productivity metrics through interactive dashboards and charts. The project demonstrates how technology can modernize traditional dairy management, improve productivity and profitability while promote digital transformation in rural and semi-urban dairy sectors. The system is scalable, reliable, and serves as a solid foundation for future enhancements.

INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 Introduction

The dairy sector is one of the most significant contributors to the economy and a primary source of livelihood for millions of people across the world. Dairy farms supply essential products such as milk, curd, butter, ghee, and ice cream to meet the daily nutritional needs of consumers. Despite its importance, many dairy farms still depend on traditional, paper-based methods for managing milk collection, farmer payments, inventory, and sales. These manual processes are prone to human errors, duplication of records, delays, and lack of transparency, which ultimately affect both farmers and customers.

The Digital Dairy Management System (DDMS) is designed to overcome these challenges by providing an automated, web-based solution for managing dairy operations. The application, developed using PHP and MySQL, offers different portals for administrators, farmers, and customers. Each user role has dedicated functionalities that help streamline the overall workflow. For example, administrators can manage farmers, track deliveries, update inventory, and view reports, while farmers can record daily milk collections and view their payments. Similarly, customers can browse products, place orders, and track deliveries online.

The system also provides real-time insights and analytics through interactive dashboards and charts, helping administrators make informed decisions. Built with responsive design principles using Bootstrap, it ensures accessibility from desktops, tablets, and mobile devices. By digitalizing dairy operations, the project promotes accuracy, efficiency, and transparency in the supply chain, while also improving the experience of farmers and customers alike.

1.2 Problem Statement

Dairy farms play a vital role in the economy, but many of them face major operational challenges due to outdated record-keeping practices. Milk collection details are often written down manually, which can result in missing or inaccurate data. This leads to disputes between farmers and dairy management over the quantity of milk submitted and the payments to be made. In addition, farmers have very limited means to check their historical submissions or monitor their earnings in real time. Customers also face difficulties because dairy farms without digital solutions often lack online product catalogues, e-commerce options, and delivery tracking systems. They cannot place orders conveniently or check the status of their deliveries. On the other side, administrators spend a lot of time updating farmer records, product stock, and sales registers without proper automation tools. Generating analytical reports for business growth is even more difficult because the data is spread across multiple registers or spreadsheets. The absence of a centralized system for managing farmers, customers, milk collection, inventory, and sales results in inefficiency, errors, and delays. Therefore, the problem this project aims to address is the lack of an integrated, digital solution for dairy farm

management that can streamline operations, improve accuracy, and increase transparency among all stakeholders.

1.3 Scope and Relevance of the Project

The scope of the Digital Dairy Management System (DDMS) covers the entire cycle of operations in a dairy farm. The project is built as a web-based application so that it can be accessed anytime and from anywhere within the network. It integrates the three main user groups—administrators, farmers, and customers—into a single platform while maintaining role-based access.

For administrators, the system provides full control over managing farmers, tracking milk collections, updating products, processing sales, generating reports, and scheduling deliveries. They can monitor overall dairy performance through dashboards and charts that display key statistics such as milk collected, revenue earned, and sales trends.

For farmers, the system offers a personalized portal where they can record their daily milk submissions. They can also view their payment history, which ensures transparency and reduces conflicts over earnings.

For customers, the system functions as a digital shopping portal where they can view available dairy products, add them to a shopping cart, place orders, and track delivery status. This makes the dairy products more accessible and improves customer satisfaction.

The relevance of this project lies in its ability to modernize dairy operations and eliminate the inefficiencies of manual systems. It enhances transparency between administrators, farmers, and customers by providing accurate and up-to-date information. It also creates a strong foundation for future growth by allowing easy scalability, integration with mobile applications, and possible expansion to larger cooperative models. For academic purposes, the project demonstrates how traditional industries can benefit from modern digital technologies.

1.4 Objectives

The primary objective of this project is to design and develop a Digital Dairy Management System that automates the core functions of dairy farm operations. The project is guided by the following specific objectives:

1. **Automation of Records:** To replace manual record-keeping by automating milk collection records and farmer payment calculations.
2. **Role-Based Access:** To provide a secure login and authentication system for administrators, farmers, and customers.
3. **Product and Inventory Management:** To enable administrators to manage dairy products, update stock quantities, and handle inventory efficiently.

4. **Online Sales and Ordering:** To implement a customer portal that supports product browsing, shopping cart, checkout, and delivery tracking.
5. **Data Visualization:** To design an interactive administrator dashboard with charts, statistics, and reports for better decision-making.
6. **Data Security and Integrity:** To ensure that user data remains consistent and protected through the use of UUID-based identifiers and secure authentication methods.
7. **Responsive Design:** To create a mobile-friendly web application that can be accessed conveniently across devices.
8. **Transparency and Efficiency:** To foster trust between farmers, customers, and administrators by providing accurate, real-time data and reducing manual errors.

SYSTEM ANALYSIS

CHAPTER 2

SYSTEM ANALYSIS

2.1 Introduction

System analysis is a critical phase in software development that involves studying the existing processes, understanding stakeholder requirements, identifying limitations, and designing an optimal solution. For the Digital Dairy Management System (DDMS), this phase is particularly important because traditional dairy farm operations involve numerous manual processes that are prone to errors and inefficiencies.

The objective of system analysis in this project is to:

- Understand the limitations of existing manual or semi-digital systems
- Identify key requirements of administrators, farmers, and customers
- Propose a computerized solution that improves accuracy, efficiency, and transparency
- Assess the feasibility of implementing the system from technical, operational, and economic perspectives

This chapter provides a detailed examination of the existing system, highlights its shortcomings, and introduces the proposed web-based solution, outlining its advantages and feasibility.

2.2 Existing System

Most dairy farms, especially small to medium-sized ones, rely heavily on manual methods for managing operations. The current system typically involves:

Manual Record Keeping:

- Milk collection records maintained in physical notebooks or basic spreadsheets
- Farmer payment calculations done manually using calculators
- Inventory tracking through paper-based stock registers
- Sales records written in daily ledger books

Communication Methods:

- Customer orders received via telephone calls or in-person visits
- Delivery schedules managed through verbal communication
- Farmer notifications through physical notice boards or word-of-mouth

Data Management:

- Separate records maintained for different operations (milk collection, sales, inventory)
- No integration between various operational data
- Physical storage of records vulnerable to damage and loss

2.2.1 Limitations of Existing System

The manual dairy management system suffers from several significant limitations:

1. Data Inaccuracy and Errors

- Manual data entry leads to calculation errors in milk quantities and payments
- Transcription mistakes during record transfer between different registers
- Difficulty in maintaining consistent data across multiple record books

2. Time-Consuming Operations

- Daily milk collection recording requires extensive manual effort
- Payment calculations for each farmer performed individually
- Report generation involves manual aggregation of data from multiple sources
- Inventory updates require physical stock counting and manual entry

3. Lack of Transparency

- Farmers cannot verify their milk collection history or payment calculations
- Customers have no visibility into order status or delivery tracking
- Limited access to historical data for performance analysis

4. Inefficient Communication

- Delayed information flow between farmers, administrators, and customers
- No real-time updates on milk collection, inventory levels, or order status
- Dependency on physical presence for information access

5. Limited Accessibility

- Records stored in physical location only
- No remote access capability for managers or stakeholders
- Single point of access creates operational bottlenecks

6. Poor Data Security

- Physical records vulnerable to damage, theft, or loss
- No backup mechanisms for critical business data
- Lack of access controls for sensitive information

7. Inadequate Reporting and Analysis

- Manual compilation of reports is time-consuming and error-prone
- Limited analytical capabilities for business intelligence
- Difficulty in identifying trends and patterns
- No real-time dashboard for operational monitoring

8. Scalability Issues

- Manual processes become increasingly difficult as business grows
- Adding new farmers or products increases administrative burden exponentially
- Limited capacity to handle increased transaction volumes

2.3 Proposed System

The Digital Dairy Management System (DDMS) is a comprehensive web-based solution designed to overcome the limitations of the existing manual system. The proposed system centralizes all dairy farm operations into a single, integrated platform with the following key features:

Centralized Database Architecture

- Unified database storing all operational data
- Real-time data synchronization across all modules
- Secure data backup and recovery mechanisms

Role-Based Access Control

- Separate interfaces for administrators, farmers, and customers
- Customized dashboards based on user roles
- Secure authentication and authorization mechanisms

Automated Operational Modules

- **Milk Collection Management:** Digital recording of daily milk collections with quality parameters

- **Farmer Management:** Comprehensive farmer profiles with payment tracking
- **Inventory Management:** Real-time stock monitoring and automated alerts
- **Sales and Order Management:** Online ordering system with order tracking
- **Delivery Management:** Route optimization and delivery status tracking
- **Payment Processing:** Automated payment calculations and transaction records

Advanced Reporting and Analytics

- Interactive dashboards with real-time metrics
- Customizable reports with multiple export options
- Data visualization using charts and graphs
- Trend analysis and performance indicators

2.3.1 Advantages of Proposed System

The DDMS offers numerous advantages over the traditional manual system:

1. Enhanced Accuracy and Reliability

- Automated calculations eliminate human errors
- Data validation rules ensure information integrity
- Consistent data entry through standardized forms

2. Improved Operational Efficiency

- Automated routine tasks reduce manual workload
- Quick data retrieval and processing
- Simultaneous multi-user access to system

3. Real-Time Information Access

- Instant access to current milk collection data
- Real-time inventory updates
- Live order and delivery tracking

4. Increased Transparency

- Farmers can view their collection history and payments online
- Customers can track order status in real-time
- Administrators have complete visibility into operations

5. Better Decision Support

- Comprehensive analytics for business intelligence
- Performance tracking and trend analysis
- Data-driven decision making capabilities

6. Enhanced Security

- Role-based access control protects sensitive data
- Automated data backups prevent information loss
- Audit trails for all system activities

7. Scalability and Flexibility

- Easy accommodation of business growth
- Modular architecture allows future enhancements
- Support for multiple locations and users

8. Cost Effectiveness

- Reduced administrative costs through automation
- Minimized errors leading to financial savings
- Improved resource utilization

2.4 Feasibility Study

A comprehensive feasibility study was conducted to evaluate the practicality of implementing the Digital Dairy Management System from technical, operational, and economic perspectives.

2.4.1 Technical Feasibility

The technical feasibility assessment confirms that the DDMS can be successfully implemented using current technology:

Technology Stack Compatibility

- **Backend:** PHP 8.0+ provides robust server-side processing capabilities
- **Database:** MySQL 8.0+ offers reliable data storage and management
- **Frontend:** HTML5, CSS3, JavaScript, and Bootstrap ensure responsive user interfaces
- **Server Environment:** Apache web server with XAMPP/WAMP stack for easy deployment

Development Resources

- Availability of skilled developers in PHP and MySQL
- Extensive documentation and community support for chosen technologies
- Compatibility with existing hardware infrastructure

Performance Considerations

- Efficient database design supports expected transaction volumes
- Responsive design ensures compatibility with various devices
- Modular architecture allows performance optimization

Integration Capabilities

- RESTful API support for future integrations
- Standard data formats for reporting and exports
- Compatibility with common web browsers

2.4.2 Operational Feasibility

The operational feasibility assessment evaluates how well the system fits into daily dairy operations:

User Acceptance

- Intuitive user interface requires minimal training
- Role-based access matches existing organizational structure
- Gradual transition plan from manual to digital systems

Workflow Integration

- Digital processes align with existing operational workflows
- Automated reporting replaces manual compilation
- Real-time data access improves decision-making speed

Training Requirements

- Minimal training needed for computer-literate users
- Comprehensive user documentation and help systems
- Phased implementation approach for smooth transition

Maintenance Considerations

- Web-based architecture simplifies updates and maintenance

- Modular design allows independent module updates
- Clear maintenance procedures and documentation

2.4.3 Economic Feasibility

The economic feasibility analysis demonstrates the financial viability of the DDMS:

Development Costs

- **Software Tools:** Open-source technologies (PHP, MySQL, Bootstrap) eliminate licensing costs
- **Hardware:** Utilizes existing computer infrastructure with minimal upgrades required
- **Development Time:** Efficient development using modern frameworks and tools

Operational Cost Savings

- **Reduced Labor Costs:** Automation of manual tasks reduces administrative workload
- **Error Reduction:** Minimized financial losses due to calculation errors
- **Improved Efficiency:** Faster operations lead to cost savings in time and resources

Tangible Benefits

- **Increased Revenue:** Online ordering expands customer base and sales
- **Better Inventory Management:** Reduced stockouts and wastage
- **Improved Farmer Relations:** Transparent payments enhance trust and loyalty

Intangible Benefits

- **Enhanced Decision Making:** Data-driven insights improve business strategies
- **Competitive Advantage:** Digital transformation positions business for future growth
- **Stakeholder Satisfaction:** Improved service quality for farmers and customers

Return on Investment

- Payback period estimated within first year of operation
- Significant long-term cost savings through automation
- Scalable solution supports business growth without proportional cost increase

2.5 SOFTWARE ENGINEERING PARADIGM APPLIED

This establishment and use of sound engineering principles in order to obtain economically developed software that is reliable and works efficiently on real machines is called software engineering

Software engineering is the discipline whose aim is:

1. Production of quality products.
2. Product that is delivered on time.
3. Cost within the budget.
4. Satisfies all requirements.

Software process is the way in which we produce the software. Apart from hiring smart, knowledgeable engineers and buying the latest development tools, effective software development process is also needed, so that engineers can systematically use the best technical and managerial practices to successfully complete their projects.

A **software life cycle** is the series of identifiable stages that a software product undergoes during its lifetime. A software lifecycle model is a descriptive and diagrammatic representation of the software life cycle. A life cycle model represents all the activities required to make a software product transit to life cycle phases. It also captures the order in which these activities are to be taken.

Life Cycle Models

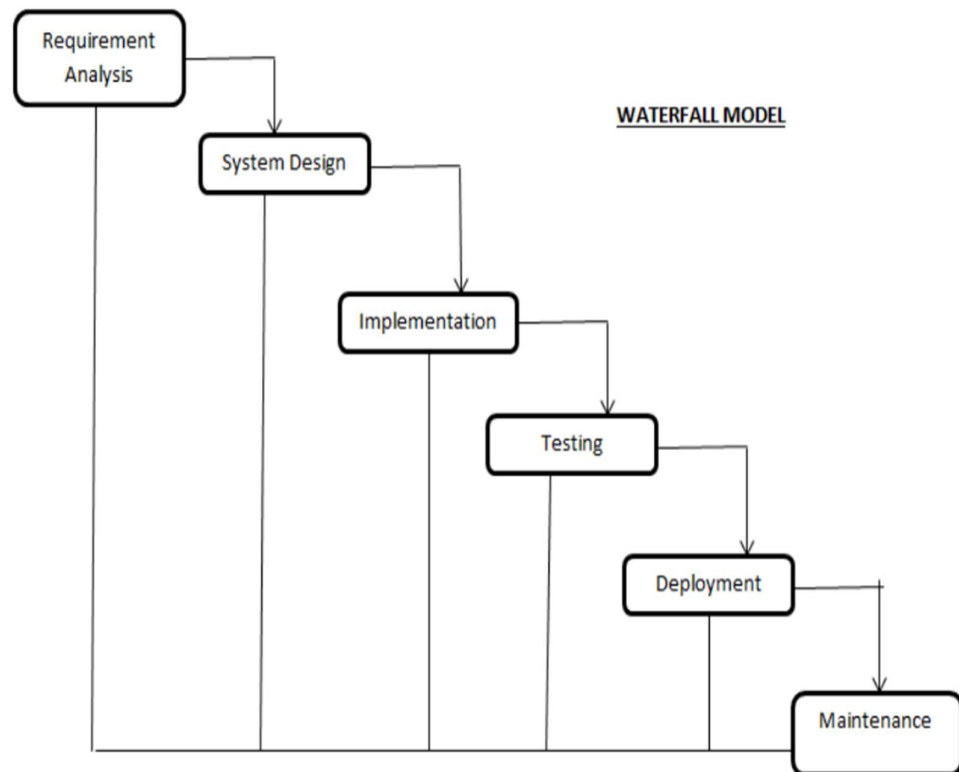
There are various Life Cycle Models to improve the software processes

1. Waterfall Model
2. Prototype Model
3. Iterative Enhancement Model
4. Evolutionary Model
5. Spiral Model

In this project, the waterfall model is followed.

The Waterfall approach was the first SDLC Model to be used widely in software engineering to ensure success of the project. In “The Waterfall” approach, the whole process of software development is divided into separate phrases. In this Waterfall model, typically, the outcome of one phrase acts as the input for the next phase sequentially.

The following illustration is a representation of the different phases of the Waterfall Model:



SYSTEM DESIGN

CHAPTER 3

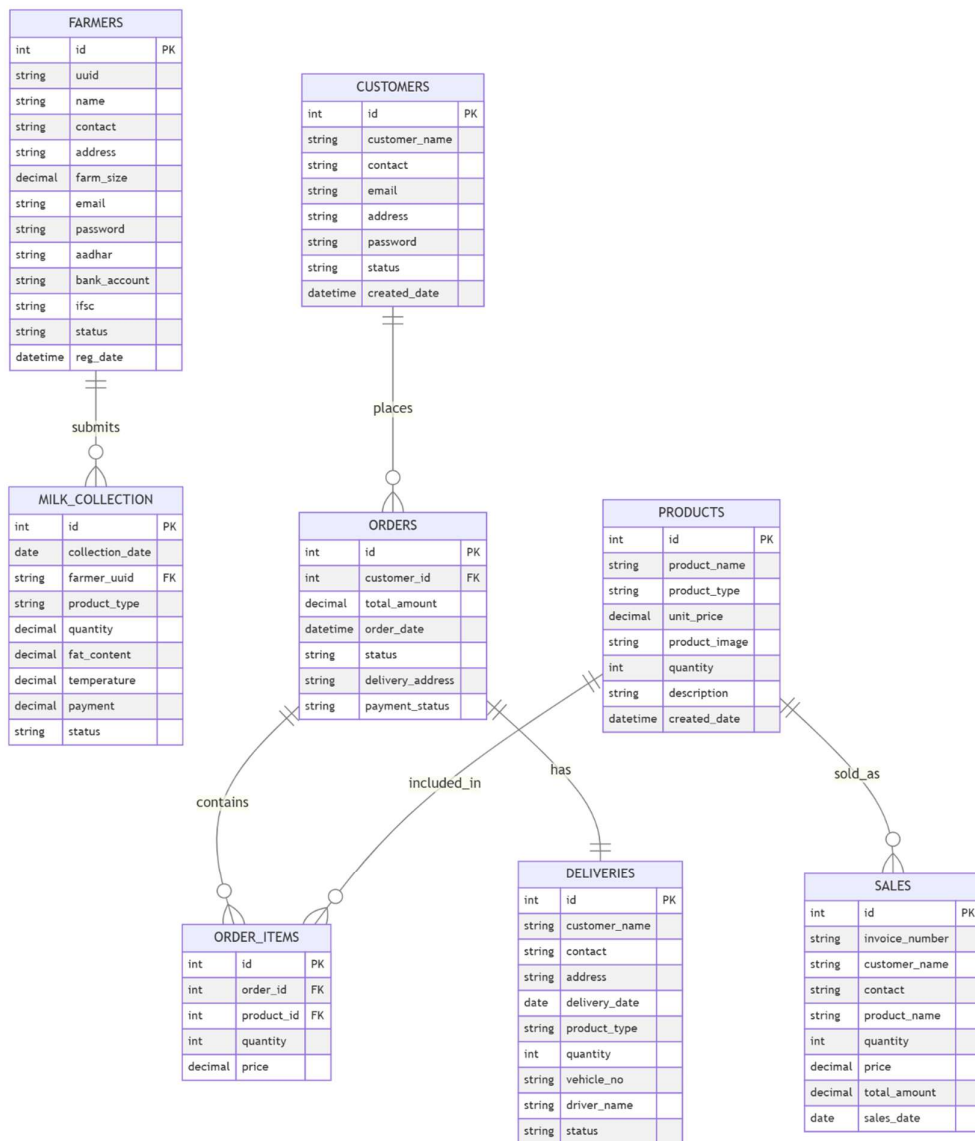
SYSTEM DESIGN

3.1 INTRODUCTION

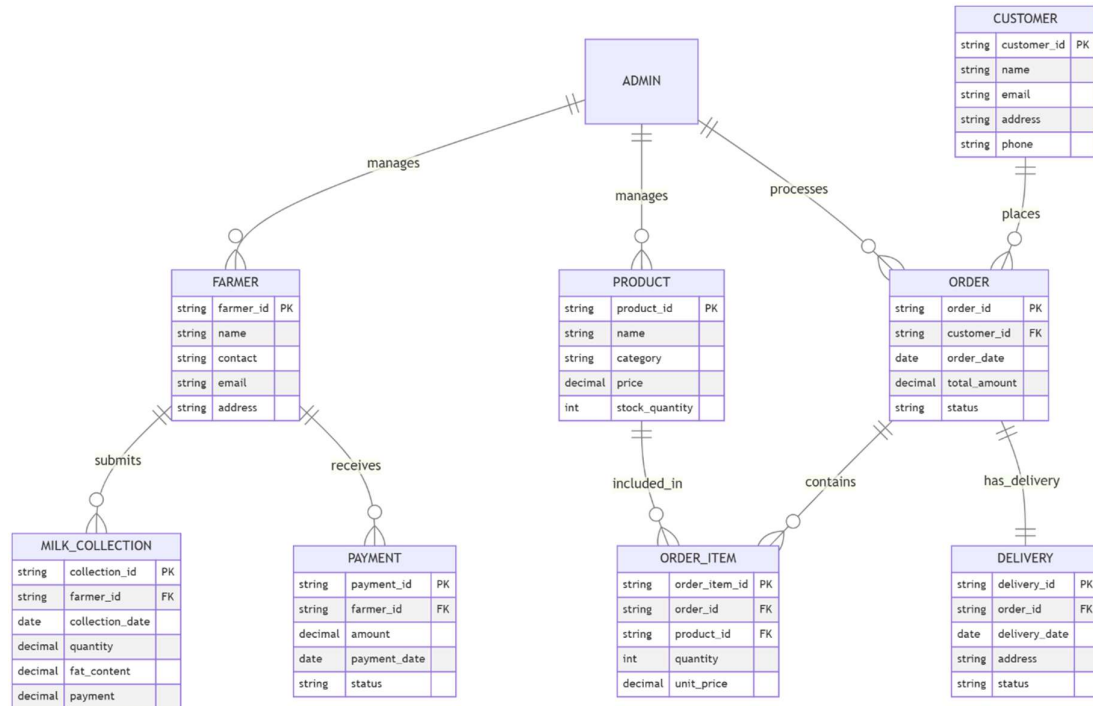
System design transforms abstract requirements into a concrete blueprint for the Digital Dairy Management System. This chapter provides comprehensive design specifications covering database architecture, process flows, object-oriented models, and user interface design to ensure efficient, scalable, and user-friendly system implementation.

3.2 DATABASE DESIGN

Database Schema Architecture



3.2.1 ENTITY RELATIONAL MODEL

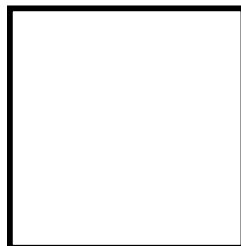


3.3 PROCESS DESIGN - DATA FLOW DIAGRAMS

A data flow diagram (DFD) is a graphical representation of the “flow” of data through an information system, modelling its process aspects. Often, they are a preliminary step used to create an overview of the system which can later be elaborated. DFDs shows what kind of information will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of the processes, or information about whether processes will operate in sequence or in parallel.

DFD SYMBOLS

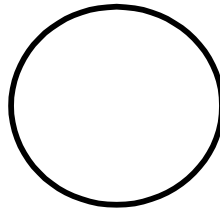
1. A **SQUARE** defines a source or destination of system data



An **ARROW** identifies data flow or data in motion. It is a pipeline through which information flows.



2. A **CIRCLE or a BUBBLE** represents a process transforms incoming data flow into outgoing data flow.



3. An **OPEN RECTANGLE** is a data store or data at rest or a temporary test repository of data



Note that the DFD describes what data flows rather than how they are processed, so it does not depend on hardware, software and data structure or file organization.

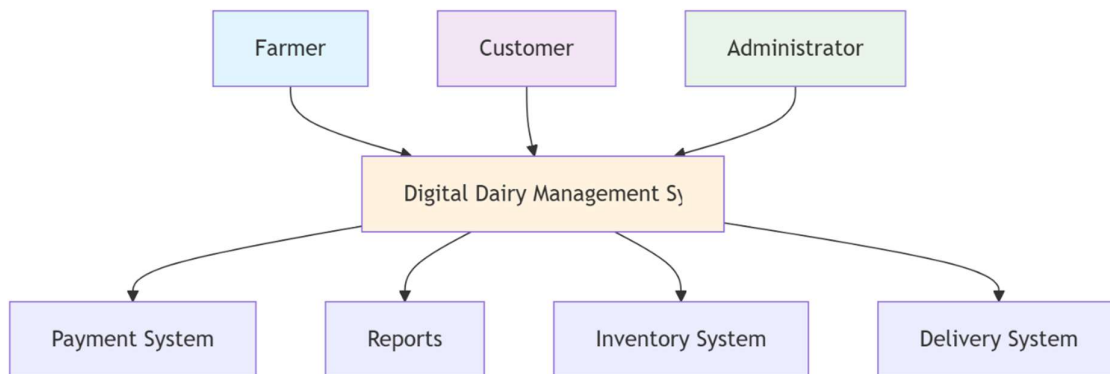
One of the tools of structured analysis is the diagram. A data flow diagram is a graphical representation of the system. The analyst can use the data flow diagram to explain this understanding about the system.

1. Data flows are an initiative way of showing how data is processed by a system
2. Data flow models are used to show how data flows through a sequence of processing steps.

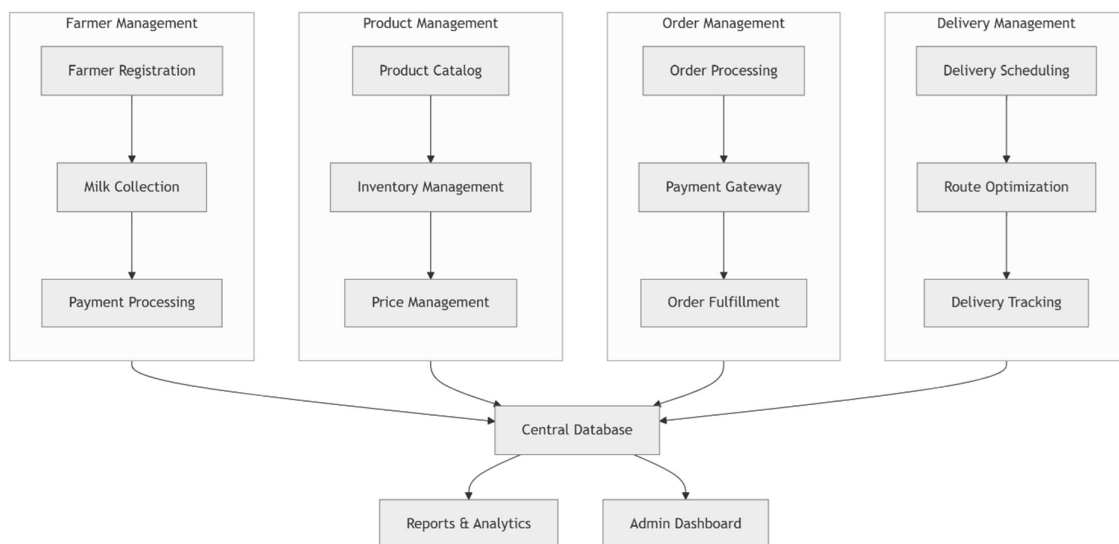
Four steps are commonly used to construct a DFD:

1. Processes should be named and numbered for easy references.
2. The direction of flow is from top to bottom and from left to right.
3. When a process is exploded into lower-level details, they are numbered.
4. The name of data stores, sources and destinations are written in capital letters.

Context Diagram (Level 0 DFD)



Level 1 DFD - Core Processes



3.4 OBJECT ORIENTED DESIGN - UML DIAGRAMS

Object-Oriented Design (OOD) is a design methodology that emphasizes organizing a system around objects—entities that encapsulate both data (attributes) and behaviour (methods). This approach closely mirrors real-world modelling, making it intuitive for developers, analysts, and stakeholders to understand and implement.

In the context of the Digital Dairy Administration System, OOD plays a vital role in structuring the system into logical components such as *Farmers*, *Milk Collection*, *Products*, *Sales*, and *Deliveries*. Each module is represented as an object that interacts with others to perform specific operations efficiently.

OOD also serves as a bridge between analysis and implementation, ensuring that all system requirements are mapped into modular, maintainable, and reusable code. It is closely integrated with Unified Modelling Language (UML), which provides a standardized way to visualize the architecture and flow of the system.

Key Aspects of OOD in UML and Database Design

UML Diagrams: UML provides a standardized way to visualize the design of a system. Several types of UML diagrams are particularly relevant to OOD and database design:

- 1. Class Diagrams:** These diagrams represent the static structure of the system by showing classes, their attributes, methods, and relationships (associations, inheritance, etc.). Class diagrams are essential for defining the data model and understanding how different entities interact.
- 2. Object Diagrams:** These diagrams depict instances of classes (objects) and their relationships at a specific point in time. They help in visualizing the state of the system and can be useful for understanding complex interactions.
- 3. Use Case Diagrams:** These diagrams illustrate the functional requirements of the system by showing actors (users or other systems) and their interactions with the system. They help identify the key functionalities that the database must support.
- 4. Sequence Diagrams:** These diagrams represent the dynamic behaviour of the system by showing how objects interact over time. They are useful for understanding the flow of data and operations, which can inform database design decisions.

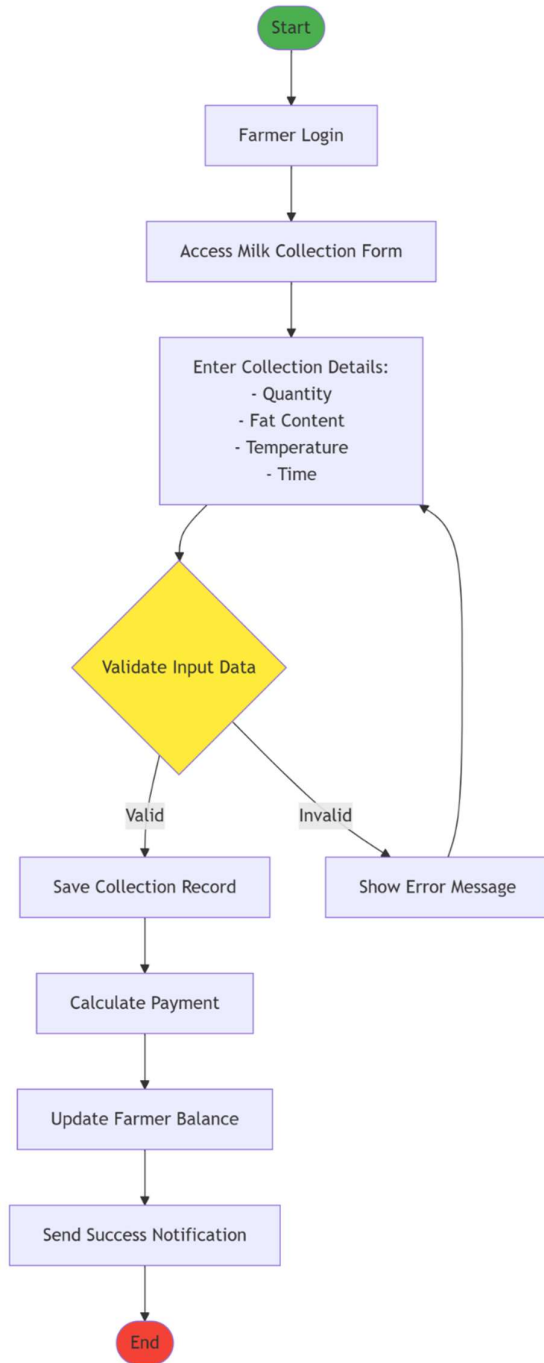
3.5 ACTIVITY DIAGRAM

An Activity Diagram represents the workflow or sequence of activities within a system. It shows how various processes are carried out step by step, the flow of control from one activity to another, and how decisions affect the overall process.

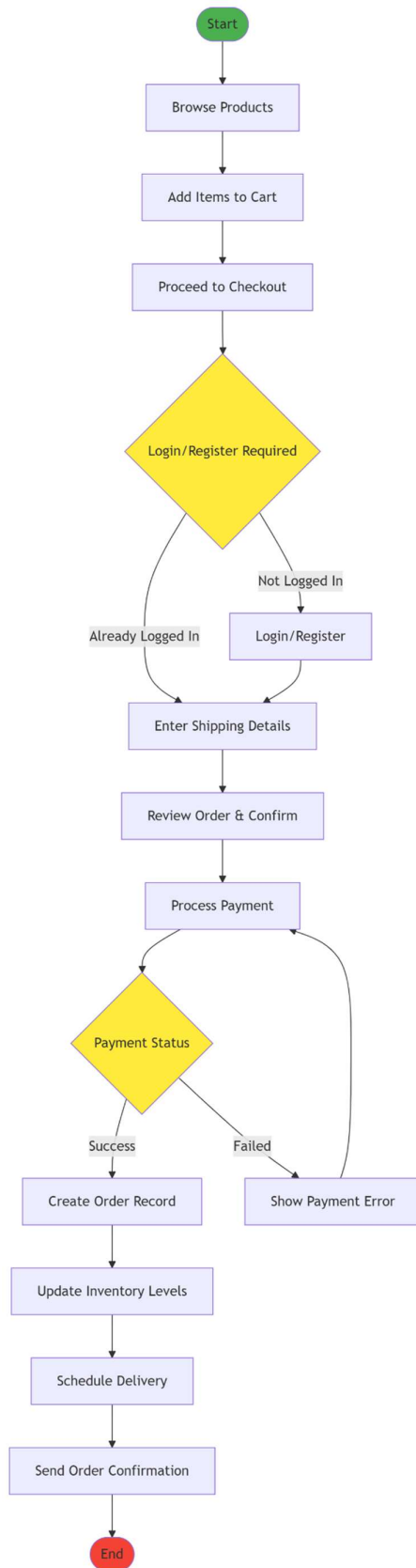
In the Digital Dairy Administration System, the activity diagram illustrates the dynamic behaviour of the system — how users such as Admin, Farmer, and Customer interact with

different modules like milk collection, product management, sales, and delivery. The main activity diagram involved in this ULM Activity Diagram of DDMS are follows:

Milk Collection Activity Flow



Order Processing Activity Flow



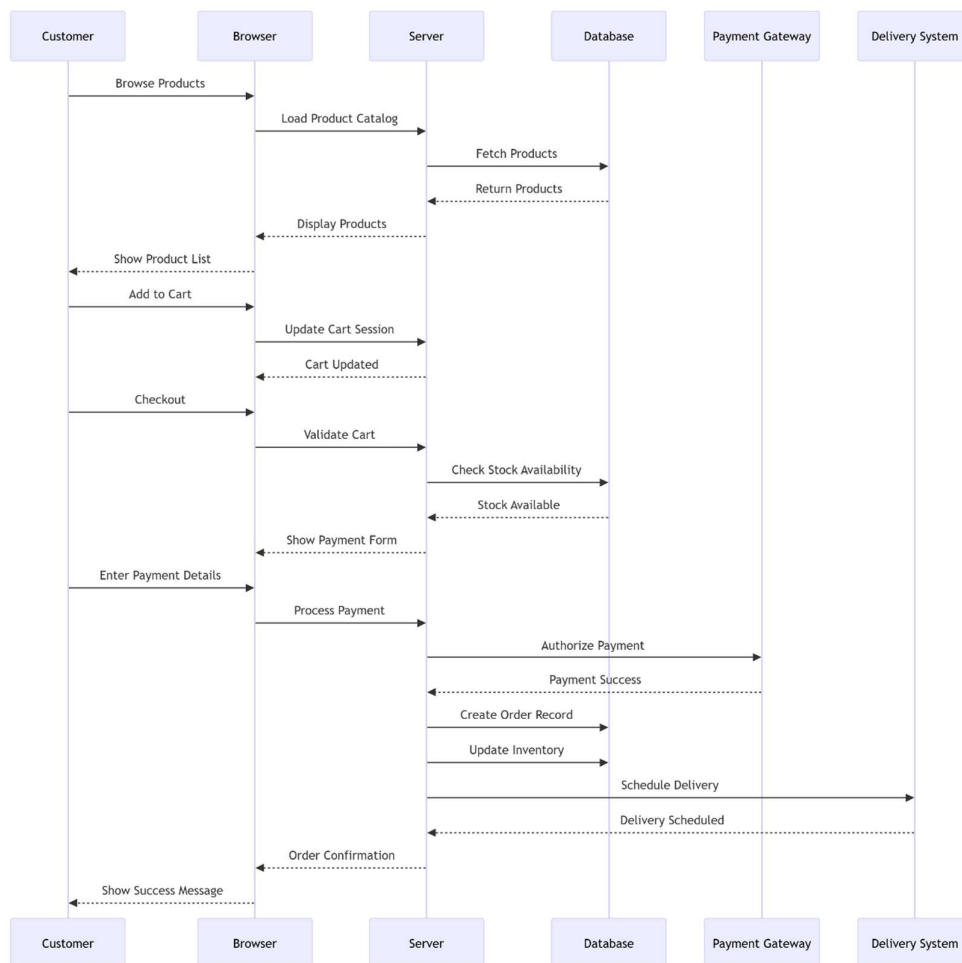
3.6 SEQUENCE DIAGRAM

Sequence diagrams describe the dynamic behaviour of the system by showing how objects communicate through messages over time. They capture the flow of interactions between system components, helping to visualize the logical order of operations — such as a farmer submitting milk data, the system recording it in the database, and the admin generating payment reports.

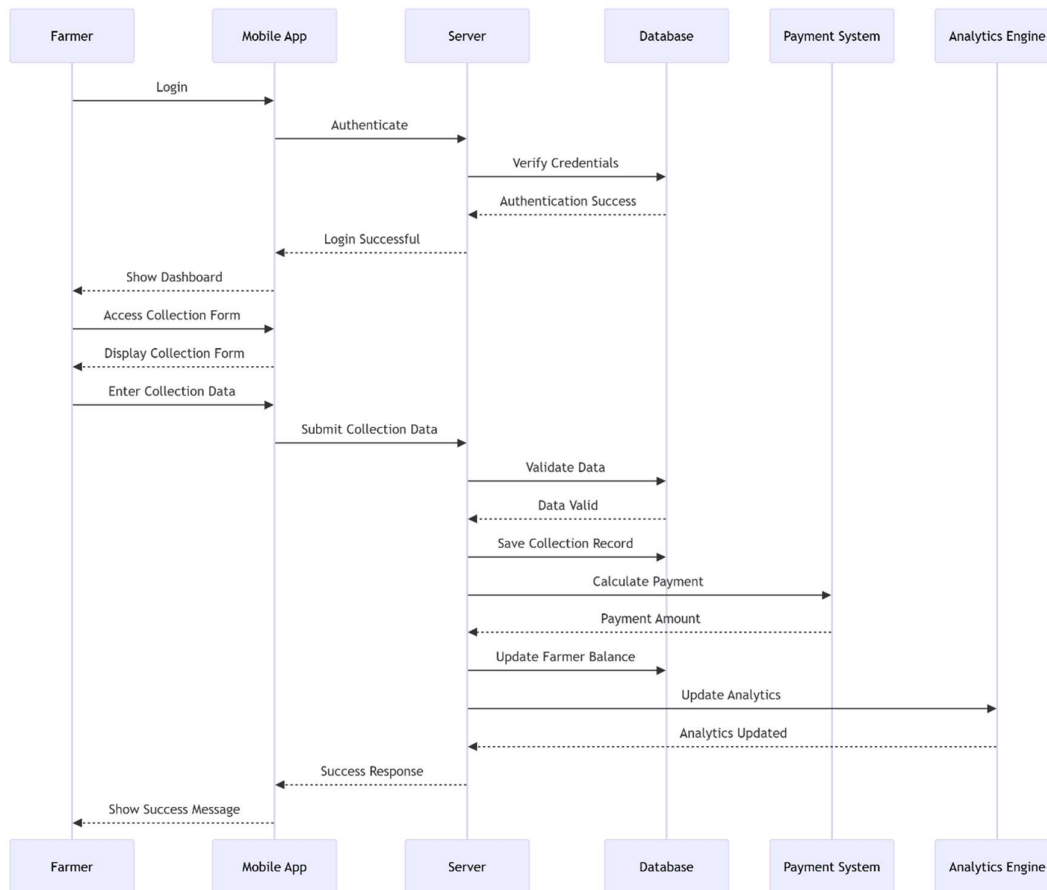
Purpose of Sequence Diagram

- Model high-level interaction between active objects in a system.
- Model the interaction between object instances within a collaboration that realizes a use case.
- Model the interaction between objects within a collaboration that realizes an operation.
- Either model generic interactions (showing all paths through the interaction) or specific instances of an interaction showing just one path through the interaction.

Customer Order Sequence



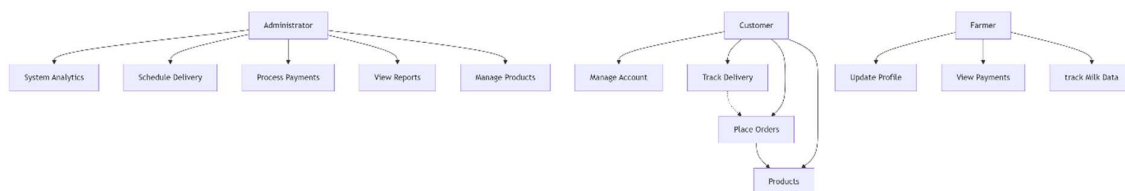
Milk Collection Sequence



3.7 USE CASE DIAGRAMS

A use case diagram in the Unified Modelling Language (UML) is a type of behavioural diagram defined by and created from a Use-case analysis. The main purpose of a use case diagram is to show what system functions are performed for which actor. A use case describes a sequence of actions that provide something of measurable value to an actor and is drawn as a horizontal ellipse. An actor is a person, organization, or external system that plays a role in one or more interactions with the system. A rectangle is drawn around the use cases, called the system boundary box, to indicate the scope of the system.

Comprehensive Use Case Overview



3.8 INPUT DESIGN

Input design is one of the most crucial phases in system design. It focuses on how data enters the system and how users interact with input screens, forms, and fields. The main goal is to ensure accurate, complete, and efficient data entry that minimizes errors and enhances user experience.

Features of Database

Compactness	Eliminates the need for large paper files by digitally storing data efficiently.
Speed	Enables rapid data retrieval and updates, much faster than manual processes.
Less Drudgery	Automates repetitive data handling tasks, reducing manual effort.
Currency of Information	Provides up-to-date and specific data as required.
Simplicity	Offers an easy way to collect, access, and display information.
Stability	Prevents unnecessary data loss through structured storage.
Security	Protects data from unauthorized access.

Primary Key

A Primary Key is a unique identifier for each record in a relational database. It ensures that each record can be uniquely retrieved or referenced. Example: Farmer ID, Product ID, or Invoice Number. A database can have only one primary key per table.

Foreign Key

A **Foreign Key** is a field (or collection of fields) in one table that uniquely identifies a row in another table. It is used to establish and enforce a link between the data in two tables. Example: In the milk collection table, farmer id is a foreign key referring to farmers.id.

Normalization

Normalization is the process of splitting relations into well-structured tables that allow users to insert, delete, and update records without introducing database inconsistencies. Without normalization, many problems can occur when trying to load an integrated conceptual model into the DBMS. These problems arise from relations that are generated directly from user views and are called anomalies. There are three types of anomalies: insertion, deletion, and update anomalies.

Anomalies in DBMS

Anomalies are problems that arise when changes are made to the database. The three major types are:

- **INSERTION Anomaly** – Occurs when certain information cannot be inserted into the database due to missing related data.
Example: When information about a new farmer cannot be added until a related milk collection record exists.
- **DELETION Anomaly** – Occurs when deleting one piece of data inadvertently removes additional useful information.
Example: Deleting a farmer's record may also remove all related milk supply data.
- **UPDATE Anomaly** – Happens when redundant data causes inconsistencies during updates.
Example: Updating a farmer's contact number in one table but not in other leads to mismatched data.

Normalization Forms

Here are the most commonly used normal forms in database design:

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce & Codd Normal Form (BCNF)

First Normal Form (1NF)

As per the rule of First Normal Form, an attribute (column) of a table cannot hold multiple values. Each column must contain only atomic (indivisible) values, and each record must be unique.

Second Normal Form (2NF)

A table is said to be in 2NF if both of the following conditions are met:

- The table is already in 1NF.
- All non-prime attributes are fully functionally dependent on the primary key.

An attribute that is not part of any candidate key is known as a **non-prime attribute**.

Third Normal Form (3NF)

A table design is said to be in 3NF if the following conditions hold:

- The table is already in 2NF.
- There is no **transitive functional dependency** between non-prime attributes and the primary key.

In other words, a table is in 3NF if for every functional dependency $X \rightarrow Y$, at least one of the following holds:

- X is a super key of the table, or
- Y is a prime attribute of the table.

Boyce & Codd Normal Form (BCNF)

BCNF is a stricter version of 3NF. A table is in BCNF if every determinant is a candidate key. It removes all remaining anomalies that are not handled by 3NF and ensures the highest level of normalization in practical systems.

TABLES

1. USER MANAGEMENT INPUT TABLES

1.1 Admin Login Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
email	VARCHAR (100)	NOT NULL	Email format validation	NULL	Admin email address
password	VARCHAR (255)	NOT NULL	Min 6 characters	NULL	Admin password

1.2 Farmer Registration Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
name	VARCHAR (100)	NOT NULL	Length: 2-100 chars, Letters only	NULL	Farmer's full name
username	VARCHAR (100)	UNIQUE	Length: 3-50 chars, Alphanumeric	NULL	Login username
email	VARCHAR (100)	UNIQUE, NOT NULL	Email format validation	NULL	Email address
contact	VARCHAR (20)	NULL	10-15 digits	NULL	Contact number
address	TEXT	NOT NULL	Length: 10-500 chars	NULL	Complete address
farm_size	DECIMAL (10,2)	NULL	Range: 0-1000 acres	0.00	Farm size in acres

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
aadhar	VARCHAR (12)	NULL	12 digits exactly	NULL	Aadhar number
bank_account	VARCHAR (30)	NULL	9-18 digits	NULL	Bank account number
ifsc	VARCHAR (15)	NULL	Pattern: ^[A-Z]{4}0[A-Z0-9]{6}\$	NULL	IFSC code
password	VARCHAR (255)	NOT NULL	Min 6 characters	NULL	Encrypted password

1.3 Customer Registration Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
customer_name	VARCHAR (255)	NOT NULL	Length: 2-255 chars	NULL	Customer's full name
contact	VARCHAR (20)	NOT NULL	10-15 digits	NULL	Contact number
email	VARCHAR (255)	UNIQUE, NOT NULL	Email format validation	NULL	Email address
address	TEXT	NULL	Length: 10-1000 chars	NULL	Delivery address
password	VARCHAR (255)	NOT NULL	Min 6 characters	NULL	Encrypted password

1.4 User Login Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
username	VARCHAR (255)	NOT NULL	Valid email or username	NULL	Username/Email
password	VARCHAR (255)	NOT NULL	Min 6 characters	NULL	Password
user_type	ENUM('farmer','customer','admin')	NOT NULL	Must be valid type	NULL	Type of user

2. MILK COLLECTION INPUT TABLES

2.1 Milk Collection Entry Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
date	DATE	NOT NULL	Not future date	CURRENT_DATE	Collection date
farmer_uuid	VARCHAR (4)	FOREIGN KEY, NOT NULL	Valid farmer UUID	NULL	Farmer reference
product_type	VARCHAR (100)	NOT NULL	Length: 2-100 chars	'milk'	Type of product
quantity	FLOAT	NOT NULL	Range: 0.1-1000.0	NULL	Quantity in liters
fat	FLOAT	NOT NULL	Range: 2.0-8.0%	NULL	Fat percentage
temperature	VARCHAR (10)	NULL	Range: 2-40°C	NULL	Temperature
payment	FLOAT	NULL	Range: 0-100000	NULL	Calculated payment

3. PRODUCT MANAGEMENT INPUT TABLES

3.1 Product Creation Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
ProductName	VARCHAR (100)	NOT NULL	Length: 2-100 chars	NULL	Product name
ProductType	VARCHAR (50)	NOT NULL	Length: 2-50 chars	NULL	Product category
UnitPrice	DECIMAL (10,2)	NOT NULL	Range: 0.01-10000.00	NULL	Selling price
productimage	VARCHAR (255)	NULL	File types: jpg,jpeg,png,gif	NULL	Product image path
Quantity	INT	NULL	Range: 0-100000	NULL	Stock quantity
Description	TEXT	NULL	Length: 0-1000 chars	NULL	Product description

4. ORDER MANAGEMENT INPUT TABLES

4.1 Order Creation Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
Email	VARCHAR (255)	FOREIGN KEY, NOT NULL	Valid customer email	NULL	Customer email
TotalAmount	DECIMAL (10,2)	NOT NULL	Range: 0.01-100000.00	NULL	Order total amount
Status	ENUM('Pending','Processing','Delivered','Cancelled')	NOT NULL	Valid status	'Pending'	Order status

4.2 Order Items Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
OrderID	VARCHAR (50)	FOREIGN KEY, NOT NULL	Valid order ID	NULL	Order reference
ProductID	VARCHAR (50)	FOREIGN KEY, NOT NULL	Valid product ID	NULL	Product reference
Quantity	INT	NOT NULL	Range: 1-1000	NULL	Quantity ordered
Price	DECIMAL (10,2)	NOT NULL	Range: 0.01 - 10000.00	NULL	Unit price
Total	DECIMAL(10,2)	NOT NULL	Calculated field	NULL	Line total

5. DELIVERY MANAGEMENT INPUT TABLES

5.1 Delivery Scheduling Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
CustomerName	VARCHAR(100)	NOT NULL	Length: 2-100 chars	NULL	Customer name
Contact	VARCHAR(20)	NOT NULL	10-15 digits	NULL	Contact number
Address	TEXT	NOT NULL	Length: 10-1000 chars	NULL	Delivery address
DeliveryDate	DATE	NOT NULL	Not past date	NULL	Scheduled delivery date
ProductType	VARCHAR(50)	NOT NULL	Length: 2-50 chars	NULL	Product type
Quantity	INT	NOT NULL	Range: 1-10000	NULL	Delivery quantity

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
VehicleNo	VARCHAR(20)	NULL	Length: 3-20 chars	NULL	Vehicle number
DriverName	VARCHAR(100)	NULL	Length: 2-100 chars	NULL	Driver name
DriverContact	VARCHAR(20)	NULL	10-15 digits	NULL	Driver contact
delivery_status	ENUM('pending','received')	NOT NULL	Valid status	'pending'	Delivery status

6. SALES MANAGEMENT INPUT TABLES

6.1 Sales Recording Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
InvoiceNumber	VARCHAR(50)	NOT NULL	Length: 3-50 chars	NULL	Invoice number
CustomerName	VARCHAR(100)	NOT NULL	Length: 2-100 chars	NULL	Customer name
Contact	VARCHAR(20)	NOT NULL	10-15 digits	NULL	Contact number
ProductName	VARCHAR(100)	NOT NULL	Length: 2-100 chars	NULL	Product name
Quantity	INT	NOT NULL	Range: 1-10000	NULL	Quantity sold
Price	DECIMAL(10,2)	NOT NULL	Range: 0.01-10000.00	NULL	Unit price
TotalAmount	DECIMAL(10,2)	NOT NULL	Calculated field	NULL	Total amount

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
SalesDate	DATE	NOT NULL	Not future date	CURRENT_DATE	Sales date

7. PASSWORD MANAGEMENT INPUT TABLES

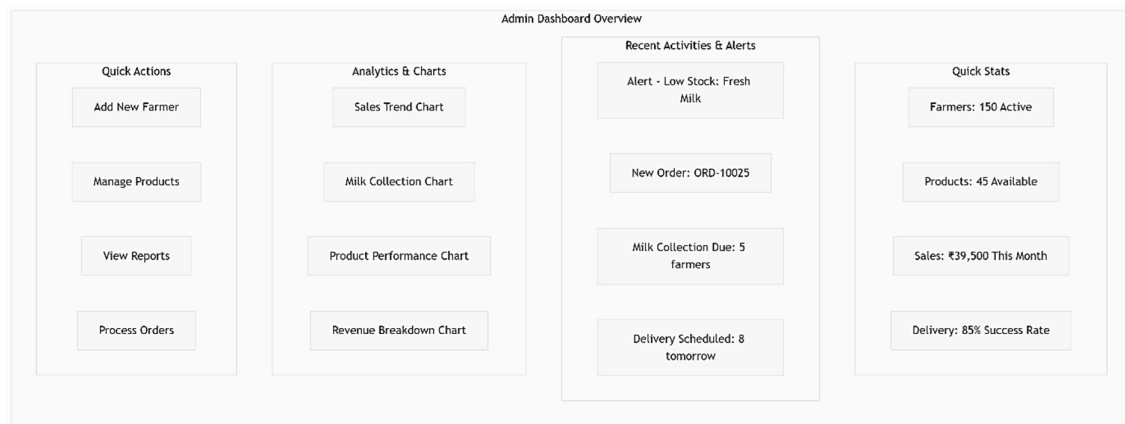
7.1 Password Reset Input Table

Field Name	Data Type	Constraints	Validation Rules	Default Value	Description
email	VARCHAR(255)	NOT NULL	Valid email format	NULL	User email
reset_token	VARCHAR(64)	NULL	Hexadecimal string	NULL	Reset token
reset_expires	DATETIME	NULL	Future datetime	NULL	Token expiry

3.9 OUTPUT DESIGN

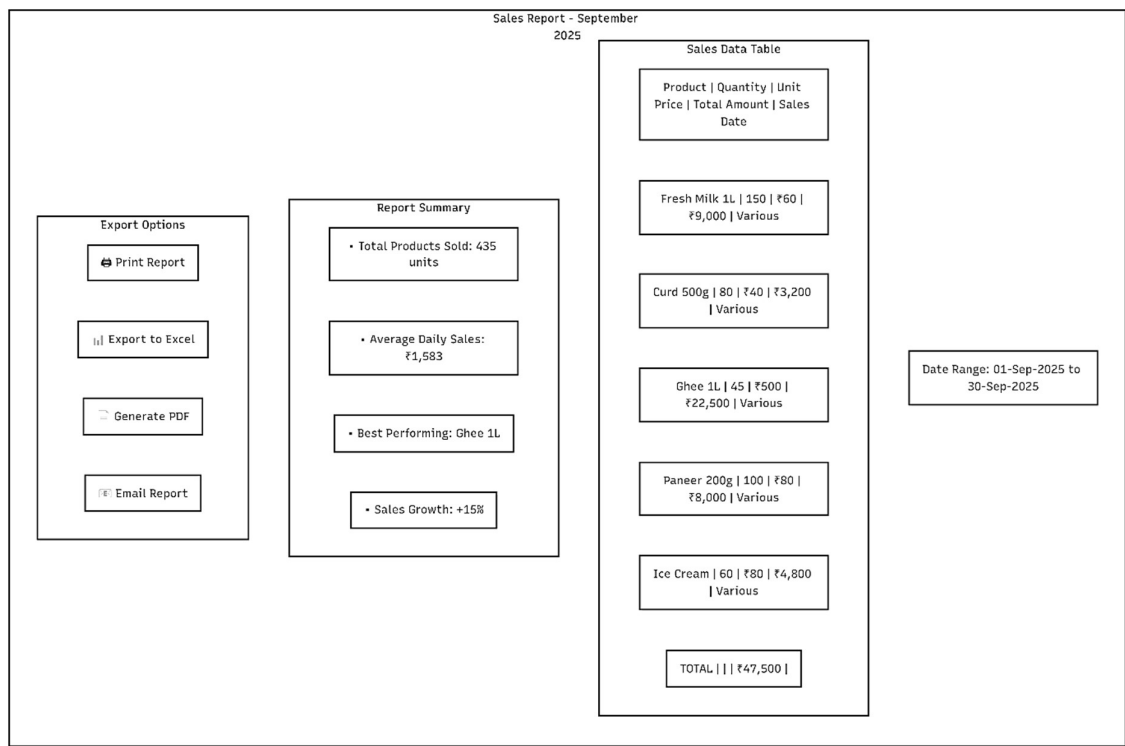
Dashboard Design

Administrator Dashboard:

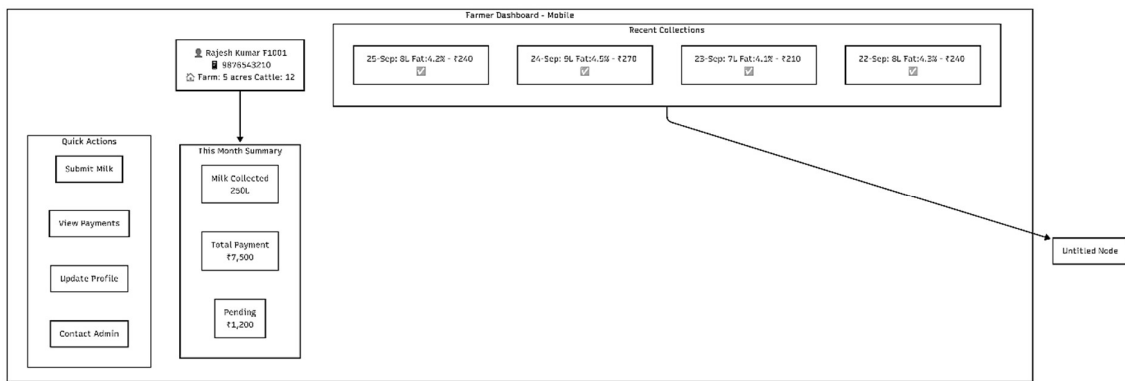


Report Design

Sales Report Output:



Farmer Dashboard:



SYSTEM ENVIRONMENT

CHAPTER 4

SYSTEM ENVIRONMENT

4.1 INTRODUCTION

The system environment refers to the combination of **hardware, software, and development tools** required for building, deploying, and running the Digital Dairy Management System (DDMS). A suitable environment ensures that the system is **efficient, reliable, scalable, and secure**.

The DDMS is a **web-based application**, so it requires:

- A **server environment** (Apache via XAMPP).
- A **database server** (MySQL) for data management.
- A **frontend environment** (HTML, CSS, JavaScript, Bootstrap) for user interface.
- A **backend environment** (PHP) for processing and business logic.

This chapter outlines the tools, platforms, and technologies used for development and execution.

4.2 TOOLS AND PLATFORMS

Hardware Requirements

- **Minimum Requirements**
 - Processor: Intel Core i3 or equivalent
 - RAM: 4 GB
 - Hard Disk: 250 GB
 - Display: 1024 × 768 resolution
- **Recommended Requirements**
 - Processor: Intel Core i5/i7 or higher
 - RAM: 8 GB or more
 - Hard Disk: 500 GB or more (SSD preferred)
 - Display: Full HD (1920 × 1080)

Software Requirements

- **Operating System:** Windows 10/11, Linux (Ubuntu), or macOS
- **Server Environment:**
 - **XAMPP** (includes Apache server, MySQL, and PHP)
- **Backend Language:** PHP 7+ (for server-side logic and database interaction)
- **Database:** MySQL (for relational data storage and management)
- **Frontend:**
 - HTML5 (page structure)
 - CSS3 + Bootstrap 5 (responsive design and styling)
 - JavaScript + Chart.js (dynamic interactions and chart visualizations)
- **IDE/Editors:**
 - Visual Studio Code
 - Sublime Text / Notepad++ (alternatives)
- **Browser:** Google Chrome / Mozilla Firefox (for testing and running the application)

Development Tools

- **phpMyAdmin** → GUI tool for managing MySQL database.
- **Git/GitHub** → Version control and project hosting.
- **Draw.io / StarUML** → Designing ER diagrams, DFDs, UML diagrams.
- **MS Word / LaTeX** → Documentation preparation.
- **MS PowerPoint** → For project presentation.

Platforms for Deployment

- **Local Deployment:**
 - XAMPP on local machine for development and testing.
- **Cloud Deployment (Optional for future):**
 - Web hosting platforms like **cPanel**, **AWS**, or **Heroku** for production deployment.

SYSTEM IMPLEMENTATION

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 INTRODUCTION

The implementation phase is the process of transforming the approved design and system architecture into an operational and functioning software product. It is the stage where theoretical designs are translated into actual executable programs using appropriate programming languages and tools.

For the **Digital Dairy Management System (DDMS)**, the implementation was carried out using **PHP** as the backend scripting language, **MySQL** as the database management system, and **Bootstrap 5** for creating a responsive and interactive user interface.

The project follows a **modular development approach**, dividing the system into several core modules such as:

- **Authentication System**
- **Farmer Management**
- **Milk Collection Management**
- **Product and Inventory Management**
- **Customer and Order Management**
- **Sales and Delivery Management**
- **Reporting and Analytics**

Each module was coded, tested, and integrated systematically to ensure proper functionality and reliability.

5.2 CODING

Coding is a vital part of system implementation. It involves converting the designed logic and algorithms into syntax understood by the computer. In this project, each module was implemented using **PHP (procedural style)** for simplicity and performance.

The development environment included:

- **Editor:** Visual Studio Code
- **Server:** Apache (via XAMPP)
- **Database:** MySQL (via phpMyAdmin)

- **Frontend:** HTML5, CSS3, Bootstrap, JavaScript
- **Charting Library:** Chart.js
- **Tables and Reports:** DataTables (with export options)

The entire codebase is organized into reusable components:

- includes/config.php – Database connection
- includes/header.php, sidebar.php, footer.php – Layout structure
- add_*.php, manage_*.php – CRUD modules for admin operations
- user_*.php – Farmer and customer panel functions
- uploads/ – File storage for product images

This modular coding strategy allows the system to be scalable and maintainable for future enhancements.

5.2.1 SAMPLE CODE

Connection (config.php)

```
<?php
$host = "localhost";
$user = "root";
$pass = "";
$dbname = "digital_dairy";
$con = mysqli_connect($host, $user, $pass, $dbname);
if (!$con) {
    die("Connection failed: " . mysqli_connect_error());
}
?>
```

Home page(home_page.php)

```
<?php
session_start();
include('includes/config.php');
$isLoggedIn = isset($_SESSION['email']);
$cartCount = isset($_SESSION['cart']) ? count($_SESSION['cart']) : 0;
```

```

?>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1">

<title>Digital Dairy Management System</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">

<link href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.5.0/css/all.min.css" rel="stylesheet">

<style>

body {font-family:Poppins,sans-serif;background:#f8f9fa;}

.hero {background:linear-gradient(90deg,#007bff,#00a8ff);color:#fff;text-align:center;padding:80px 20px;}

.category-card {background:#fff;border-radius:15px;padding:20px;box-shadow:0 3px 6px rgba(0,0,0,0.1);transition:.3s;}

.category-card:hover {transform:scale(1.05);}

.category-card img {width:100%;border-radius:10px;height:150px;object-fit:cover;}

footer {background:#fff;padding:15px 0;text-align:center;box-shadow:0 -1px 5px rgba(0,0,0,0.1);}

</style>

</head>

<body>

<nav class="navbar navbar-expand-lg bg-white shadow-sm fixed-top">

  <div class="container">

    <a class="navbar-brand fw-bold text-primary" href="#">DDMS</a>

    <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#nav"><span class="navbar-toggler-icon"></span></button>

    <div class="collapse navbar-collapse" id="nav">

      <ul class="navbar-nav ms-auto">

        <li class="nav-item"><a href="#" class="nav-link active">Home</a></li>

        <li class="nav-item"><a href="#" class="nav-link">Products</a></li>

        <li class="nav-item"><a href="#" class="nav-link">Contact</a></li>

      </ul>

      <div class="ms-3">

        <?php if($isLoggedIn): ?>

```

```

        <a href="profile.php" class="btn btn-outline-primary btn-sm"><i class="fa fa-user"></i>
        <?=$_SESSION['email']?></a>

        <a href="logout_cus.php" class="btn btn-primary btn-sm">Logout</a>

        <?php else: ?>

        <a href="index.php" class="btn btn-primary btn-sm">Login</a>

        <?php endif; ?>

    </div> </div> </div> </nav>

<div style="height:80px"></div>

<section class="hero">

    <h1>Welcome to <b>Digital Dairy</b></h1>

    <p>Smart solutions for milk collection, farmers, and products.</p>

    <a href="#products" class="btn btn-light text-primary fw-bold mt-3">Explore Products</a>

</section>

<div class="container my-5" id="products">

    <div class="text-center mb-4">

        <h2 class="fw-bold">Our Product Categories</h2> </div>

    <div class="row g-4 text-center">

        <div class="col-md-3 col-6"><div class="category-card"><h6 class="mt-2">Milk</h6></div></div>

        <div class="col-md-3 col-6"><div class="category-card"><h6 class="mt-2">Curd</h6></div></div>

        <div class="col-md-3 col-6"><div class="category-card"><h6 class="mt-2">Ice Cream</h6></div></div>

        <div class="col-md-3 col-6"><div class="category-card"><h6 class="mt-2">Ghee</h6></div></div>

    </div>

</div> </div>

<footer>

    <p class="mb-0">&copy; 2025 Digital Dairy | Developed by Joyal</p>

</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>

</body>

</html>

```

Admin login(login.php)

```
<?php
session_start();
include('includes/config.php');
if(isset($_POST['login'])){
    $email=$_POST['email']; $pass=$_POST['password'];
    if($email=='admin@gmail.com' && $pass=='admin'){
        $_SESSION['admin_id']=0; $_SESSION['email']=$email;
        header("Location: dashboard.php"); exit; }
    $q=mysqli_prepare($con,"SELECT ID,Password FROM admin WHERE email=?");
    mysqli_stmt_bind_param($q,"s",$email);
    mysqli_stmt_execute($q);
    $r=mysqli_stmt_get_result($q);
    $row=mysqli_fetch_assoc($r);
    if($row && md5($pass)===$row['Password']){
        $_SESSION['admin_id']=$row['ID']; $_SESSION['email']=$email;
        header("Location: dashboard.php"); exit;
    } else $error="Invalid email or password!"; }
?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Admin Login - DDMS</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
<link href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.5.0/css/all.min.css" rel="stylesheet">
</head>
<body class="bg-light d-flex justify-content-center align-items-center vh-100">
<form method="POST" class="bg-white p-4 rounded shadow" style="width:350px;">
    <h4 class="text-center mb-3 text-primary"><i class="fa fa-user-shield me-2"></i>Admin Login</h4>
    <?php if(!empty($error)) echo "<p class='text-danger small'>$error</p>"; ?>
    <input type="email" name="email" class="form-control mb-3" placeholder="Email" required>
    <input type="password" name="password" id="pw" class="form-control mb-3" placeholder="Password"
    required>
    <button type="button" onclick="togglePw()" class="btn btn-sm btn-link text-secondary p-0 mb-2">Show
    Password</button>
```

```

<button type="submit" name="login" class="btn btn-primary w-100">Sign In</button>

<p class="text-center mt-3"><a href="index.php" class="text-decoration-none">Back to User Login</a></p>
</form>
<script>
function togglePw(){var p=document.getElementById("pw");p.type=p.type=="password"? "text": "password";}
</script> </body>
</html>

```

Dashboard(user_dashboard.php)

```

<?php
include('includes/config.php');
include('includes/sidebar.php');

$total_farmers=20;$total_deliveries=8;$total_milk=1250;$total_revenue=45000;
?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Dashboard - Digital Dairy</title>
<link rel="stylesheet" href="style.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
</head>
<body>
<div class="dashboard">
<header>
<h1>Dashboard Overview</h1>
<p>Welcome back! Here's your dairy summary.</p>
<div class="date"><i class="fas fa-calendar"></i> <?php echo date("M d, Y"); ?></div>
</header>
<div class="stats">
<div class="card"><i class="fas fa-users"></i><h3><?php echo $total_farmers;
?></h3><p>Farmers</p></div>
<div class="card"><i class="fas fa-truck"></i><h3><?php echo $total_deliveries;
?></h3><p>Deliveries</p></div>
<div class="card"><i class="fas fa-tint"></i><h3><?php echo $total_milk; ?> L</h3><p>Milk
Collected</p></div>

```



```

        <div class="card"><i class="fas fa-rupee-sign"></i><h3>₹<?php echo $total_revenue;
?></h3><p>Revenue</p></div>

</div>

<section class="charts">

    <div><h3>Monthly Milk Collection</h3><canvas id="milkChart"></canvas></div>

    <div><h3>Product Category</h3><canvas id="pieChart"></canvas></div>

</section>

<section class="recent">

    <h3>Recent Activity</h3>

    <div class="activity"><i class="fas fa-user-plus"></i> New Farmer Added</div>

    <div class="activity"><i class="fas fa-tint"></i> 250L Milk Collected</div>

    <div class="activity"><i class="fas fa-truck"></i> Delivery Scheduled</div>

</section>

</div>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<script>new
Chart(document.getElementById('milkChart'), {type:'line',data: {labels:['Jan','Feb','Mar','Apr'],datasets:[ {label:'M
ilk(L)',data:[200,300,250,400]}]} });

new
Chart(document.getElementById('pieChart'), {type:'pie',data: {labels:['Milk','Ghee','Butter'],datasets:[ {data:[60,2
5,15]}]} });

</script> </body>

</html>

```

Farmer registration (add_farmer.php)

```

<?php

session_start();

if(!isset($_SESSION['email'])) {header("Location: login.php");exit;}

include('includes/header.php');

include('includes/config.php');

include('includes/uuid_helper.php');

$farmer_uuid = generateShortUUID();

?>

<!DOCTYPE html>

<html lang="en">

<head>

```

```

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Add Farmer - Digital Dairy</title>

<link rel="stylesheet" href="style.css">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">

</head>

<body>

<div class="dashboard-container">

    <?php include('includes/sidebar.php'); ?>

    <div class="dashboard-main">

        <div class="dashboard-header">

            <div>

                <h1>Add New Farmer</h1>

                <p>Register a new farmer in the system</p>

            </div>

            <div><i class="fas fa-calendar"></i> <span id="currentDateTime"></span></div>

        </div>

        <div class="form-card">

            <h3><i class="fas fa-user-plus"></i> Farmer Registration</h3>

            <form action="process_farmer.php" method="post" class="simple-form">

                <input type="hidden" name="farmer_uuid" value="<?php echo $farmer_uuid; ?>">

                <div class="form-grid">

                    <input type="text" name="name" placeholder="Full Name" required>

                    <input type="text" name="username" placeholder="Username" required>

                    <input type="email" name="email" placeholder="Email" required>

                    <input type="password" name="password" placeholder="Password" required>

                </div>

                <button type="submit" class="btn btn-primary">Register Farmer</button>

            </form>

        </div> </div> </div>

    </body>

</html>

<?php include('includes/footer.php'); ?>

```

Milk collection management(manage_collection.php)

```
<?php
session_start();
if(!isset($_SESSION['email'])){header("Location: login.php");exit;}
include('includes/header.php');
include('includes/config.php');
?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Manage Milk Collection - Digital Dairy</title>
<link rel="stylesheet" href="style.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
</head>
<body>
<div class="dashboard-container">
<?php include('includes/sidebar.php'); ?>
<div class="dashboard-main">
<div class="dashboard-header">
<h1>Milk Collection</h1>
<div><i class="fas fa-calendar"></i> <?php echo date('M d, Y'); ?></div>
</div>
<div class="table-box">
<h3><i class="fas fa-tint"></i> Collection Records</h3>
<table class="table" border="1" cellpadding="6" cellspacing="0">
<tr>
<th>ID</th><th>Date</th><th>Farmer</th><th>Product</th>
<th>Qty (L)</th><th>Fat%</th><th>Temp °C</th><th>Payment</th><th>Action</th>
</tr>
<?php
    $q=mysqli_query($con,"SELECT mc.*,f.name AS farmer FROM milk_collection mc JOIN farmers f ON
mc.farmer_uuid=f.uuid ORDER BY mc.date DESC");
    $i=1;
    while($r=mysqli_fetch_assoc($q)){ ?>
<tr>
<td><?= $i++; ?></td>
```

```

<td><?= date('M j, Y',strtotime($r['date'])); ?></td>
<td><?= htmlspecialchars($r['farmer']); ?></td>
<td><?= $r['product_type']; ?></td>
<td><?= number_format($r['quantity'],2); ?></td>
<td><?= number_format($r['fat'],2); ?></td>
<td><?= number_format($r['temperature'],1); ?></td>
<td><?= number_format($r['payment'],2); ?></td>
<td><a href="delete-collection.php?id=<?= $r['id']; ?>" onclick="return confirm('Delete this
record?');"><i class="fas fa-trash text-danger"></i></a></td>
</tr>
<?php } ?>
</table> </div> </div>
</body>
</html>
<?php include('includes/footer.php'); ?>

```

Delivery management (manage_delivery.php)

```

<?php
session_start();
if (!isset($_SESSION['email'])) {
    header("Location: login.php");
    exit;
}
include('includes/header.php');
include('includes/config.php');
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Manage Deliveries - Digital Dairy System</title>
<link rel="stylesheet" href="style.css">
</head>

```

```

<body>

<div class="dashboard-container">

<?php include('includes/sidebar.php'); ?>

<div class="dashboard-main">

    <?php if(isset($_SESSION['message'])): ?>

        <div class="alert alert-info"><?= htmlspecialchars($_SESSION['message']); ?></div>

        <?php unset($_SESSION['message']); endif; ?>

    <div class="chart-card">

        <div class="chart-header">

            <h3><i class="fas fa-truck"></i> All Deliveries</h3>

            <p class="text-muted">View all delivery records</p>

        </div>

        <div class="table-container table-responsive">

            <table class="table table-striped table-hover">

                <thead class="table-dark">

                    <tr>

                        <th>ID</th><th>Customer</th><th>Contact</th><th>Address</th>

                        <th>Date</th><th>Product</th><th>Qty</th><th>Vehicle</th>

                        <th>Driver</th><th>D.Contact</th><th>Posted</th><th>Status</th><th>Action</th>

                    </tr>

                </thead>

                <tbody>

                    <?php

                        $res = mysqli_query($con, "SELECT * FROM tbldelivery ORDER BY ID DESC");

                        $i = 1;

                        while ($r = mysqli_fetch_assoc($res)): ?>

                            <tr>

                                <td><?= $i++ ?></td>

                                <td><i class="fas fa-user-circle"></i> <?= htmlspecialchars($r['CustomerName']) ?></td>

                                <td><?= htmlspecialchars($r['Contact']) ?></td>

                                <td><?= htmlspecialchars(substr($r['Address'], 0, 40)) ?>...</td>

                                <td><?= date('M j, Y', strtotime($r['DeliveryDate'])) ?></td>

                                <td><?= htmlspecialchars($r['ProductType']) ?></td>


```

```

<td><?= htmlspecialchars($r['Quantity']) ?></td>

<td><?= htmlspecialchars($r['VehicleNo']) ?></td>

<td><?= htmlspecialchars($r['DriverName']) ?></td>

<td><?= htmlspecialchars($r['DriverContact']) ?></td>

<td><small><?= date('M j, Y H:i', strtotime($r['PostingDate'])) ?></small></td>

<td>

    <?php

        $s = $r['delivery_status'];

        echo $s == 'pending' ? '<span class="badge bg-warning text-dark">Pending</span>' :

            ($s == 'received' ? '<span class="badge bg-success">Received</span>' :

                '<span class="badge bg-secondary">Unknown</span>');

    ?>

</td>

<td>

    <a href="delete-delivery.php?id=<?= $r['ID']; ?>"

        class="btn btn-sm btn-danger"

        onclick="return confirm('Delete this record?');">

        <i class="fas fa-trash"></i>

    </a>    </td>    </tr>    <?php endwhile; ?>    </tbody>    </table>    </div>    </div>

</div> </div>

<?php include('includes/footer.php'); ?>

</body>

</html>

```

Product management(manage_product.php)

```

<?php

session_start();

if (!isset($_SESSION['email'])) {

    header("Location: login.php");

    exit;

}

include('includes/header.php');

include('includes/config.php');

?>

```

```

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1">

<title>Manage Products - Digital Dairy</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="dashboard-container">

<?php include('includes/sidebar.php'); ?>

<div class="dashboard-main">

  <div class="dashboard-header">

    <div class="dashboard-title">

      <h1>Manage Products</h1>

      <p>View, edit, or delete inventory products</p>

    </div>

    <div class="date-display"><i class="fas fa-calendar"></i> <span id="currentDateTime"></span></div>

  </div>

  <div class="chart-card">

    <div class="chart-header">

      <h3><i class="fas fa-box"></i> Product List</h3>

      <p class="text-muted">All available products in the system</p>

    </div>

    <div class="table-responsive">

      <table class="table table-striped table-hover">

        <thead class="table-dark">

          <tr>

            <th>ID</th><th>Image</th><th>Name</th><th>Type</th>

            <th>Price</th><th>Added</th><th>Action</th>

          </tr>

```

```

</thead>

<tbody>

<?php
$res = mysqli_query($con, "SELECT * FROM tblproduct ORDER BY ID DESC");
$i = 1;
while ($r = mysqli_fetch_assoc($res)): ?>

    <tr>

        <td><?=$i++ ?></td>

        <td>

            <?php if (!empty($r['ProductImage']) && file_exists($r['ProductImage'])): ?>

                "

                    class="product-thumbnail"

                    onclick="showImageModal('<?=$htmlspecialchars($r['ProductImage']) ?>','<?=$
htmlspecialchars($r['ProductName']) ?>')">

            <?php else: ?>

                <div class="no-image"><i class="fas fa-image"></i></div>

            <?php endif; ?>

        </td>

        <td><strong><?=$htmlspecialchars($r['ProductName']) ?></strong></td>

        <td><span class="badge bg-info"><?=$htmlspecialchars($r['ProductType']) ?></span></td>

        <td><strong class="text-success"><i class="fas fa-rupee-sign"></i> <?=$
number_format($r['UnitPrice'],2) ?></strong></td>

        <td><small class="text-muted"><?=$date('M j, Y', strtotime($r['PostingDate'])) ?></small></td>

        <td>

            <a href="edit-product.php?pid=<?=$r['ID'] ?>" class="btn btn-sm btn-warning"><i class="fas
fa-edit"></i></a>

            <a href="delete-product.php?pid=<?=$r['ID'] ?>" class="btn btn-sm btn-danger"
onclick="return confirm('Delete this product?');"><i class="fas fa-trash"></i></a>

        </td>

    </tr>

<?php endwhile; ?>

</tbody>

</table>

```



```

        </div> </div> </div> </div>

<?php include('includes/footer.php'); ?>

</body>

</html>

```

Manage sales(manage_sales.php)

```

<?php
session_start();

if (!isset($_SESSION['email'])) { header("Location: login.php"); exit; }

include('includes/header.php');
include('includes/config.php');
?>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1">

<title>Manage Sales - Digital Dairy</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="dashboard-container">

<?php include('includes/sidebar.php'); ?>

<div class="dashboard-main">

    <div class="dashboard-header">

        <div class="dashboard-title">

            <h1>Manage Sales</h1>

            <p>View and manage all sales transactions</p>

        </div>

        <div class="date-display"><i class="fas fa-calendar"></i> <span id="currentDateTime"></span></div>

    </div>

    <div class="chart-card">

        <div class="chart-header">

            <h3><i class="fas fa-table"></i> Sales Records</h3>

            <p class="text-muted">All recorded sales in the system</p>

        </div>

```

```

<div class="table-responsive">
  <table class="table table-striped table-hover">
    <thead class="table-dark">
      <tr>
        <th>ID</th><th>Invoice</th><th>Customer</th><th>Contact</th>
        <th>Product</th><th>Qty</th><th>Price</th><th>Total</th>
        <th>Date</th><th>Actions</th> </tr>
    </thead>
    <tbody>
      <?php
        $q = mysqli_query($con, "SELECT * FROM tblsales ORDER BY SalesDate DESC");
        $i = 1;
        while ($r = mysqli_fetch_assoc($q)): ?>
          <tr>
            <td><?= $i++ ?></td>
            <td><strong><?= htmlspecialchars($r['InvoiceNumber']) ?></strong></td>
            <td><?= htmlspecialchars($r['CustomerName']) ?></td>
            <td><?= htmlspecialchars($r['Contact']) ?></td>
            <td><?= htmlspecialchars($r['ProductName']) ?></td>
            <td><?= number_format($r['Quantity'], 2) ?> units</td>
            <td><?= number_format($r['Price'], 2) ?></td>
            <td><strong class="text-success"><?= number_format($r['TotalAmount'], 2) ?></strong></td>
            <td><?= date('M j, Y', strtotime($r['SalesDate'])) ?></td>
            <td>
              <a href="#" class="btn btn-sm btn-info"><i class="fas fa-eye"></i></a>
              <a href="#" class="btn btn-sm btn-warning"><i class="fas fa-edit"></i></a>
              <a href="delete-sale.php?id=<?= $r['ID'] ?>" class="btn btn-sm btn-danger" onclick="return
confirm('Delete this record?');"><i class="fas fa-trash"></i></a>
            </td> </tr>
          <?php endwhile; ?>
        </tbody>
      </table> </div>
    </div>
  </div>
</div>
<?php include('includes/footer.php'); ?>
</body> </html>

```

Customer management(manage_customer.php)

```
<?php
session_start();
if (!isset($_SESSION['email'])) { header("Location: login.php"); exit; }
include('includes/config.php');
include('includes/header.php');
if (isset($_GET['id'])) {
    $id = mysqli_real_escape_string($con, $_GET['id']);
    if ($_GET['action'] == 'activate') {
        mysqli_query($con, "UPDATE tblcustomer SET status='active' WHERE id='$id'");
        $msg = "Customer activated successfully!";
    } elseif ($_GET['action'] == 'deactivate') {
        mysqli_query($con, "UPDATE tblcustomer SET status='inactive' WHERE id='$id'");
        $msg = "Customer deactivated successfully!";
    } elseif (isset($_GET['delete'])) {
        mysqli_query($con, "DELETE FROM tblcustomer WHERE id='$id'");
        $msg = "Customer deleted successfully!";
    }
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Manage Customers</title>
<link rel="stylesheet" href="style.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
</head>
<body>
<div class="dashboard-container">
<?php include('includes/sidebar.php'); ?>
<div class="dashboard-main">
<h1>Manage Customers</h1>
<?php if(!empty($msg)) echo "<div class='success-message'><i class='fas fa-check-circle'></i> $msg</div>"; ?>
<table class="table table-hover">
<thead><tr>
```

```

<th>ID</th><th>Name</th><th>Contact</th><th>Email</th>
<th>Address</th><th>Date</th><th>Status</th><th>Action</th>

</tr>

</thead>

<tbody>

<?php
$res = mysqli_query($con, "SELECT * FROM tblcustomer ORDER BY registration_date DESC");
while($r = mysqli_fetch_assoc($res)){

$badge = $r['status']=='active' ? "<span class='badge bg-success'>Active</span>" : "<span class='badge bg-secondary'>Inactive</span>";

?>

<tr>

<td><?=$r['id'] ?></td>

<td><b><?=$r['customer_name'] ?></b></td>

<td><?=$r['contact'] ?></td>

<td><?=$r['email'] ? : 'N/A' ?></td>

<td><?=$r['address'] ?>...</td>

<td><?=$r['registration_date'] ?></td>

<td><?=$badge ?></td>

<td>

<a href="?action=<?=$r['status']=='active'?deactivate':activate' ?>&id=<?=$r['id'] ?>" class="btn btn-sm <?=$r['status']=='active'?btn-warning':btn-success' ?>">

<i class="fas fa-toggle-<?=$r['status']=='active'?off':on' ?>"></i>

</a>

<a href="?delete&id=<?=$r['id'] ?>" onclick="return confirm('Delete this customer?')" class="btn btn-sm btn-danger"><i class="fas fa-trash"></i></a>

</td>

</tr>

<?php } ?>

</tbody>

</table>

</div>

</div>

<?php include('includes/footer.php'); ?>

</body>

</html>

```

Milk collection report page(milk_collection_report.php)

```
<?php
include('includes/config.php');
?>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1.0">

<title>Milk Collection Report - Digital Dairy</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<?php include('includes/sidebar.php'); ?>

<div class="content" id="mainContent">

<h1><i class="fas fa-tint"></i> Milk Collection Report</h1>

<form method="POST" class="form-row">

  <label>From: <input type="date" name="from_date" required value="<?= $_POST['from_date'] ?? "
?>"></label>

  <label>To: <input type="date" name="to_date" required value="<?= $_POST['to_date'] ?? " ?>"></label>

  <label>Product:

    <select name="product_type">

      <option value="">All</option>

      <?php foreach(['Cow Milk','Buffalo Milk','Goat Milk'] as $p): ?>

        <option value="<?= $p ?>" <?= (($_POST['product_type']??)==$p)?'selected':" ?>"><?= $p ?></option>

      <?php endforeach; ?>

    </select>

  </label>

  <button type="submit" name="search">Search</button>

</form>

<?php if(isset($_POST['search'])):

  $from=$_POST['from_date']; $to=$_POST['to_date']; $p=$_POST['product_type']??;

  $sql="SELECT mc.*, f.name farmer FROM milk_collection mc JOIN farmers f ON f.uuid=mc.farmer_uuid
WHERE mc.date BETWEEN '$from' AND '$to'.($p?" AND mc.product_type='$p'":"")." ORDER BY mc.date
DESC";

  $res=mysqli_query($con,$sql);

?>
```

```

<table>
<thead><tr><th>#</th><th>Date</th><th>Farmer</th><th>Product</th><th>Qty
(L)</th><th>Fat%</th><th>Temp °C</th><th>Payment ₹</th></tr></thead>

<tbody>

<?php
$no=1; $tq=$tp=0;
while($r=mysqli_fetch_assoc($res)){
    $tq+=$r['quantity']; $tp+=$r['payment'];
    echo "<tr>

<td>$no</td>

<td>".date('d M Y',strtotime($r['date']))."</td>

<td>{$r['farmer']}</td>

<td>{$r['product_type']}</td>

<td>".number_format($r['quantity'],2)."</td>

<td>{$r['fat']}</td>

<td>{$r['temperature']}</td>

<td>₹".number_format($r['payment'],2)."</td>

</tr>";

    $no++;
}

echo "
        <tr><th
        colspan='4'>Total</th><th>".number_format($tq,2)."</th><th
        colspan='2'></th><th>₹".number_format($tp,2)."</th></tr>";
?>

</tbody></table>

<?php endif; ?>

</div>

</body>

</html>

```

Login page(index.php)

```

<?php
session_start();
include('includes/config.php');
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $login = mysqli_real_escape_string($con, $_POST['login_input']);
    $pass = md5($_POST['password']);

```

```

$type = $_POST['user_type'];
if ($type === 'farmer') {
    $field = filter_var($login, FILTER_VALIDATE_EMAIL) ? 'email' : 'username';
    $sql = "SELECT * FROM farmers WHERE $field='$login' AND password='$pass' LIMIT 1";
} elseif ($type === 'customer') {
    $sql = "SELECT * FROM tblcustomer WHERE email='$login' AND password='$pass' LIMIT 1";
}
$res = mysqli_query($con, $sql);
if ($res && mysqli_num_rows($res) === 1) {
    $u = mysqli_fetch_assoc($res);
    $_SESSION['email'] = $u['email'];
    $_SESSION['name'] = $u['name'] ?? $u['customer_name'];
    $_SESSION['user_type'] = $type;
    $_SESSION['farmer_uuid'] = $u['uuid'] ?? null;
    header("Location: " . ($type === 'farmer' ? 'user_dashboard.php' : 'home_page.php'));
    exit;
} else $error = "Invalid credentials or user type.";
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1.0">
<title>Login - Digital Dairy</title>
<link rel="stylesheet" href="style.css">
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.4.0/css/all.min.css">
</head>
<body>
<div class="login-container">
<h2><i class="fas fa-sign-in-alt"></i> Login</h2>
<?php if(isset($error)) echo "<p class='error'>$error</p>"; ?>
<form method="POST">
<select name="user_type" required>
<option value="">Select User</option>
<option value="farmer">Farmer</option>
<option value="customer">Customer</option>

```

```

</select>

<input type="text" name="login_input" placeholder="Username or Email" required>

<input type="password" name="password" placeholder="Password" required>

<button type="submit">Sign In</button>

</form>

<p>New user? <a href="customer_reg.php">Create Account</a></p>

</div>

</body>

</html>

```

farmer dashboard (user_dashboard.php)

```

<?php
session_start();

if(!isset($_SESSION['email'])) { header("Location:index.php"); exit; }

include('includes/config.php');

// Example variable placeholders — replace with actual DB queries

$farmer_name = $_SESSION['name'] ?? 'Farmer';
$farmer_uuid = $_SESSION['farmer_uuid'] ?? '';
$today_collection = $today_rate = $today_amount = $total_collection = $total_records = $total_payments = 0;
$notifications = []; // Example: fetch from DB

?>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1.0">

<title>User Dashboard - Digital Dairy</title>

<link rel="stylesheet" href="css/user_style.css">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">

</head>

<body>

<div class="dashboard-container">

<?php include('includes/user_sidebar.php'); ?>

<div class="main-content">

```



```

    <h1>Welcome,      <?=htmlspecialchars($farmer_name)?><?=      $farmer_uuid      ?      "      (UUID:
".htmlspecialchars($farmer_uuid).")" : "" ?></h1>

    <p>Overview of your milk collection and payments</p>

    <div class="stats-grid">

        <?php
        $stats = [

            ['icon'=>'fa-tint','value'=>number_format($today_collection,2).' L','label'=>"Today's Milk Collection"],
            ['icon'=>'fa-rupee-sign','value'=>'₹'.number_format($today_rate,2),'label'=>"Today's Rate"],
            ['icon'=>'fa-money-bill-wave','value'=>'₹'.number_format($today_amount,2),'label'=>"Today's Amount"],
            ['icon'=>'fa-tint','value'=>number_format($total_collection,2).' L','label'=>'Total Collection'],
            ['icon'=>'fa-chart-line','value'=>number_format($total_records),'label'=>'Total Records'],
            ['icon'=>'fa-rupee-sign','value'=>'₹'.number_format($total_payments,2),'label'=>'Total Payments']

        ];

        foreach($stats as $s): ?>

            <div class="stat-card">

                <div class="stat-icon"><i class="fas <?=$s['icon']?>"></i></div>

                <h3><?=$s['value']?></h3><p><?=$s['label']?></p>

            </div>

        <?php endforeach; ?>

    </div>

    <div class="notifications">

        <h3>Latest Collections</h3>

        <?php if(empty($notifications)): ?>

            <p>No recent collections found.</p>

        <?php else: foreach($notifications as $n): ?>

            <p><strong><?=date('d      M      Y',strtotime($n['date']))?>:</strong>      <?=$n['quantity']?>      L,
₹<?=number_format($n['payment'],2)?></p>

        <?php endforeach; endif; ?>

    </div>

</div>

</div>

</body>

</html>

```

Farmer milk collection record (user_milk_record.php)

```
<?php
session_start();
if (!isset($_SESSION['email'])) { header("Location: index.php"); exit; }
include('includes/config.php');
$from = $_GET['from-date'] ?? "";
$to = $_GET['to-date'] ?? "";
$email = $_SESSION['email'];
$q = mysqli_query($con, "SELECT * FROM farmers WHERE email='$email'");
$f = mysqli_fetch_assoc($q);
$uuid = $f['uuid'] ?? null;
$name = $f['name'] ?? $_SESSION['name'];
$summary = ['milk'=>0, 'earn'=>0, 'avg'=>0];
if ($uuid) {
    $m = mysqli_fetch_assoc(mysqli_query($con, "
        SELECT SUM(quantity) milk, SUM(payment) earn, AVG(quantity) avg
        FROM milk_collection
        WHERE farmer_uuid='$uuid' AND MONTH(date)=MONTH(CURDATE()) AND
        YEAR(date)=YEAR(CURDATE())
    "));
    if ($m) $summary = ['milk'=>$m['milk']??0, 'earn'=>$m['earn']??0, 'avg'=>$m['avg']??0];
} $where = $uuid ? "WHERE farmer_uuid='$uuid' " : "";
if ($from) $where .= " AND date>='$from'";
if ($to) $where .= " AND date<='$to'";
$records = [];
if ($uuid) {
    $r = mysqli_query($con, "SELECT * FROM milk_collection $where ORDER BY date DESC");
    while ($row = mysqli_fetch_assoc($r)) {
        $row['rate'] = $row['quantity']>0 ? $row['payment']/$row['quantity'] : 0;
        $records[] = $row + ['status'=>'Completed']; }
    $t = mysqli_fetch_assoc(mysqli_query($con, "SELECT SUM(quantity) q, SUM(payment) p, COUNT(*) c
    FROM milk_collection $where"));
    $tot = ['q'=>$t['q']??0, 'p'=>$t['p']??0, 'c'=>$t['c']??0]; }
?>

<!DOCTYPE html>
<html lang="en">
```

```

<head>

<meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1">

<title>Milk Collection Records - Digital Dairy</title>

</head>

<body>

<div class="dashboard-container">

<?php include('includes/user_sidebar.php'); ?>

<div class="main-content">

<div class="page-title">

<h1>Milk Collection Records</h1>

<a href="?download=csv?=(($from?"&from-date=$from:".").($to?"&to-date=$to:"."))?" class="btn btn-
primary">Download Report</a>

</div>

<div class="card">

<div class="card-header"><h2>Filter Records</h2></div>

<form method="GET" class="filter-group">

<div><label>From Date</label><input type="date" name="from-date" value="<?=$from?>"></div>

<div><label>To Date</label><input type="date" name="to-date" value="<?=$to?>"></div>

<div><br><button class="btn btn-primary">Apply</button>

<button type="button" onclick="location.href='user_milk_record.php'" class="btn btn-
secondary">Clear</button></div>

</form> </div>

<div class="card">

<div class="card-header">

<h2>Milk Collection History</h2>

<div><button class="btn btn-secondary" onclick="print()">Print</button>

<button class="btn btn-primary" onclick="exportPDF()">Export PDF</button></div> </div>

<table id="milkTable" class="table">

<thead><tr><th>Date</th><th>Product</th><th>Qty(L)</th><th>Fat(%)</th><th>Temp(°C)</th><th>Rate(₹/
L)</th><th>Amount(₹)</th><th>Status</th></tr></thead>

<tbody>

<?php if(!$records): ?><tr><td colspan="8" align="center">No records</td></tr>

<?php else: foreach($records as $r): ?>

<tr>

<td><?=date('d/m/Y',strtotime($r['date']))?></td>

<td><?=$r['product_type']?></td>

<td><?=number_format($r['quantity'],2)?></td>

```

```

        <td><?=number_format($r['fat'],2)?></td>
        <td><?=number_format($r['temperature'],1)?></td>
        <td>₹<?=number_format($r['rate'],2)?></td>
        <td>₹<?=number_format($r['payment'],2)?></td>
        <td><span class="status completed">Completed</span></td>
    </tr>
<?php endforeach; endif; ?>
</tbody>
</table>
</div>
<div class="stats-grid">
    <div><h3><?=number_format($tot['q'],2)?> L</h3><p>Total Collection</p></div>
    <div><h3>₹<?=number_format($tot['p'],2)?></h3><p>Total Payment</p></div>
    <div><h3><?=$tot['c']?></h3><p>Total Records</p></div>
</div>
<div class="card">
    <div class="card-header"><h2>Monthly Summary</h2></div>
    <div style="display:flex;justify-content:space-around;">
        <div><h3><?=number_format($summary['milk'],2)?> L</h3><p>Milk This Month</p></div>
        <div><h3>₹<?=number_format($summary['earn'],2)?></h3><p>Earnings</p></div>
        <div><h3><?=number_format($summary['avg'],2)?> L</h3><p>Avg/Day</p></div>
    </div> </div></div> </div> </body> </html>

```

User payment (user_payment.php)

```

<?php
session_start();
if(!isset($_SESSION['email'])) {header("Location:index.php");exit;}
include('includes/config.php');
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Payment History - Digital Dairy</title>
</head>
<body>
<div class="dashboard-container">

```

```

<?php include('includes/user_sidebar.php'); ?>
<div class="main-content">
  <div class="page-title">
    <h1>Payment History</h1>
    <form method="GET" style="display:inline;">
      <input type="hidden" name="from-date" value="<?=htmlspecialchars($from_date)?>">
      <input type="hidden" name="to-date" value="<?=htmlspecialchars($to_date)?>">
      <button type="submit" name="download_report" value="1" class="btn btn-primary">Download
Report</button>
    </form>
  </div>
  <div class="card">
    <div class="card-header"><h2>Filter Payments</h2></div>
    <form method="GET" class="filter-group">
      <div><label>From          Date</label><input          type="date"          name="from-date"
value="<?=htmlspecialchars($from_date)?>"></div>
      <div><label>To          Date</label><input          type="date"          name="to-date"
value="<?=htmlspecialchars($to_date)?>"></div>
      <div><br><button class="btn btn-primary">Apply</button>
      <button type="button" onclick="location.href='user_payment.php'" class="btn btn-
secondary">Clear</button></div>
    </form>
  </div>
  <div class="card">
    <div class="card-header">
      <h2>Payment History</h2>
    </div>
    <button class="btn btn-secondary" style="margin-right:10px;">Print</button>
    <form method="GET" style="display:inline;">
      <input type="hidden" name="export_pdf" value="1">
      <input type="hidden" name="from-date" value="<?=htmlspecialchars($from_date)?>">
      <input type="hidden" name="to-date" value="<?=htmlspecialchars($to_date)?>">
      <button class="btn btn-primary">Export PDF</button>
    </form>
  </div>
</div>
<div class="table-responsive">

```

```

<table id="paymentTable" class="table">
  <thead><tr><th>Date</th><th>Amount (₹)</th><th>Status</th></tr></thead>
  <tbody>
    <?php if(empty($payments)): ?>
      <tr><td colspan="3" align="center">No payments found.</td></tr>
    <?php else: foreach($payments as $p): ?>
      <tr>
        <td><?=date('d/m/Y',strtotime($p['date']))?></td>
        <td>₹<?=number_format($p['payment'],2)?></td>
        <td><span class="status <?=strtolower($p['status'])?>"><?=$p['status']?></span></td>
      </tr>
    <?php endforeach; endif; ?>
  </tbody>
</table>
</div>
</div>
<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon revenue"><i class="fas fa-rupee-sign"></i></div>
    <div class="stat-content"><h3>₹<?=number_format($total_payments,2)?></h3><p>Total
Payments</p></div>
  </div>
  <div class="stat-card">
    <div class="stat-icon info"><i class="fas fa-chart-line"></i></div>
    <div class="stat-content"><h3><?=number_format($total_records)?></h3><p>Total Records</p></div>
  </div> </div> </div> </div> </body> </html>

```

Profile management page(user_profile.php)

```

<?php
session_start();
if (!isset($_SESSION['email'])) { header("Location: index.php"); exit; }
include('includes/config.php');
?>
<!DOCTYPE html>
<html lang="en">
<head>

```

```

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1">

<title>My Profile - Digital Dairy</title>

<link rel="stylesheet" href="css/user_style.css">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">

</head>

<body>

<div class="dashboard-container">

    <?php include('includes/user_sidebar.php'); ?>

    <div class="main-content">

        <h1 class="page-title">My Profile</h1>

        <div class="card">

            <div class="card-header"><h2>Profile Information</h2></div>

            <div class="profile-avatar"><i class="fas fa-user-circle"></i></div>

            <?php if(isset($success)) echo "<div class='alert alert-success'>$success</div>"; ?>

            <?php if(isset($error)) echo "<div class='alert alert-error'>$error</div>"; ?>

            <form method="POST">

                <label><i class="fas fa-user"></i> Full Name *</label>

                <input type="text" name="name" value="<?php echo htmlspecialchars($farmer_name); ?>" required>

                <label><i class="fas fa-envelope"></i> Email</label>

                <input type="email" value="<?php echo htmlspecialchars($user_email); ?>" readonly>

                <label><i class="fas fa-id-badge"></i> Farmer UUID</label>

                <input type="text" value="<?php echo htmlspecialchars($farmer_uuid); ?>" readonly>

                <label><i class="fas fa-at"></i> Username</label>

                <input type="text" name="username" value="<?php echo htmlspecialchars($farmer_username); ?>">

                <label><i class="fas fa-phone"></i> Contact Number</label>

                <input type="tel" name="phone" value="<?php echo htmlspecialchars($farmer_phone); ?>">

                <label><i class="fas fa-map-marker-alt"></i> Address *</label>

                <textarea name="address" rows="3" required><?php echo htmlspecialchars($farmer_address);
?></textarea>

                <label><i class="fas fa-tractor"></i> Farm Size (acres)</label>

                <input type="number" name="farm_size" value="<?php echo htmlspecialchars($farmer_farm_size); ?>"
step="0.01" min="0">

                <label><i class="fas fa-id-card"></i> Aadhar Number</label>

                <input type="text" name="aadhar" value="<?php echo htmlspecialchars($farmer_aadhar); ?>"
maxlength="12">

                <label><i class="fas fa-university"></i> Bank Account</label>

```

```

        <input type="text" name="bank_account" value="<?php echo htmlspecialchars($farmer_bank_account);
?>">

        <label><i class="fas fa-code"></i> IFSC Code</label>

        <input type="text" name="ifsc" value="<?php echo htmlspecialchars($farmer_ifsc); ?>">

        <button type="submit" class="btn btn-primary">Update Profile</button>

    </form> </div> </div> </div> </body> </html>

```

Report page(user_report.php)

```

<?php
session_start();

if (!isset($_SESSION['email'])) { header("Location: index.php"); exit; }

include('includes/config.php');

?>

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1">

    <title>Reports - Digital Dairy</title>

    <link rel="stylesheet" href="css/user_style.css">

    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">

</head>

<body>

<div class="dashboard-container">

    <?php include('includes/user_sidebar.php'); ?>

    <div class="main-content">

        <h1 class="page-title">Milk Collection Reports</h1>

        <div class="card">

            <h2>Filter Report Period</h2>

            <form method="GET" class="filter-group">

                <div>

                    <label>From Date</label>

                    <input type="date" name="from-date" value="<?php echo htmlspecialchars($from_date); ?>">

                </div>

                <div>

                    <label>To Date</label>

```



```

        <input type="date" name="to-date" value="<?php echo htmlspecialchars($to_date); ?>">
    </div>

    <div>

        <button type="submit" class="btn btn-primary">Generate</button>

        <button type="button" onclick="location.href='user_report.php'" class="btn btn-
secondary">Clear</button>

    </div>

</form>

</div>

<div class="card">

    <h2>Summary - <?php echo htmlspecialchars($report_data['period']); ?></h2>

    <div class="report-summary">

        <div class="summary-card"><i class="fas fa-tint"></i><h3><?php
number_format($report_data['total_collection'],2); ?> L</h3><p>Total Milk</p></div>

        <div class="summary-card"><i class="fas fa-rupee-sign"></i><h3>₹<?php
number_format($report_data['total_payments'],2); ?></h3><p>Total Payments</p></div>

        <div class="summary-card"><i class="fas fa-chart-line"></i><h3><?php
number_format($report_data['total_records']); ?></h3><p>Records</p></div>

        <div class="summary-card"><i class="fas fa-calendar-day"></i><h3><?php
number_format($report_data['average_daily'],2); ?> L</h3><p>Average Daily</p></div>

    </div>

</div> <div class="card">

    <h2>Actions</h2>

    <button class="btn btn-primary" onclick="window.print()">Print</button>

    <button class="btn btn-secondary" id="exportPdfBtn">PDF</button>

    <button class="btn btn-secondary" id="exportExcelBtn">Excel</button>

</div> </div></div>

<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/jspdf.umd.min.js"></script>

<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf-
autotable/3.5.28/jspdf.plugin.autotable.min.js"></script>

<script src="https://cdnjs.cloudflare.com/ajax/libs/xlsx/0.18.5/xlsx.full.min.js"></script>

<script>

const pdfBtn=document.getElementById('exportPdfBtn');
const excelBtn=document.getElementById('exportExcelBtn');
pdfBtn.onclick=(()=>{const {jsPDF}=window.jspdf;const doc=new jsPDF();
doc.text("Milk Collection Report",14,20);
doc.text("Period: <?php echo htmlspecialchars($report_data['period']); ?>",14,30);
doc.autoTable({head:[['Metric','Value']],body:[

```

```

['Total Milk (L)','<?php echo number_format($report_data['total_collection'],2); ?>'],
['Total Payments (₹)','₹<?php echo number_format($report_data['total_payments'],2); ?>'],
['Total Records','<?php echo number_format($report_data['total_records']); ?>'],
['Average Daily (L)','<?php echo number_format($report_data['average_daily'],2); ?>'],
],startY:40});doc.save('milk_collection_report.pdf');});
excelBtn.onclick={()=>{const wb=XLSX.utils.book_new();
const data=[['Metric','Value'],
['Total Milk (L)','<?php echo number_format($report_data['total_collection'],2); ?>'],
['Total Payments (₹)','₹<?php echo number_format($report_data['total_payments'],2); ?>'],
['Total Records','<?php echo number_format($report_data['total_records']); ?>'],
['Average Daily (L)','<?php echo number_format($report_data['average_daily'],2); ?>']];
const ws=XLSX.utils.aoa_to_sheet(data);
XLSX.utils.book_append_sheet(wb,ws,"Report");
XLSX.writeFile(wb,"milk_collection_report.xlsx");}};
</script> </body> </html>

```

Milk product page(milk_product.php)

```

<?php
include('includes/header.php');
include('includes/config.php');
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Milk Products - Digital Dairy</title>
  <link rel="stylesheet" href="style.css">
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css">
</head>
<body>
<div class="full-width-container">
  <section class="hero-section">
    <div class="hero-content">
      <h1>Milk Products</h1>
      <p>Fresh, pure milk from trusted dairy farms</p>

```

```

        <a href="#products" class="btn-order-now"><i class="fas fa-shopping-cart"></i> View Products</a>
    </div> </section>

<section class="category-navigation">
    <a href="milk_product.php" class="category-link active">Milk</a>
    <a href="curd_product.php" class="category-link">Curd</a>
    <a href="ghee_product.php" class="category-link">Ghee</a>
    <a href="icecream_product.php" class="category-link">Ice Cream</a>
</section>

<section id="products" class="products-section">
    <h2>Our Milk Products</h2>
    <div class="row">
        <?php
            $ret=mysqli_query($con,"SELECT * FROM tblproduct WHERE ProductType LIKE '%Milk%' ORDER BY
            ID DESC");
            if(mysqli_num_rows($ret)>0){
                while($row=mysqli_fetch_assoc($ret)){ ?>
                    <div class="product-card">
                        <div class="product-image">
                            <?php if(!empty($row['ProductImage']) && file_exists($row['ProductImage'])) { ?>
                                "
                                <?php } else { ?><i class="fas fa-cow" style="font-size:3rem;color:#ccc;"></i><?php } ?>
                            </div>
                        <div class="product-info">
                            <h3><?php echo htmlspecialchars($row['ProductName']); ?></h3>
                            <p><?php echo htmlspecialchars($row['ProductType']); ?></p>
                            <strong>₹<?php echo number_format($row['UnitPrice'],2); ?></strong>
                            <form method="POST" action="cart.php">
                                <input type="hidden" name="action" value="add">
                                <input type="hidden" name="product_id" value="<?php echo $row['ID']; ?>">
                                <input type="hidden" name="quantity" value="1">
                                <button class="btn-add-to-cart"><i class="fas fa-shopping-cart"></i> Add</button>
                            </form>
                        </div>
                    </div>
                <?php }
            } else { ?>
                <div class="no-products"><i class="fas fa-cow"></i><h3>No Milk Products Available</h3></div>
            <?php } ?>
        </div>
    </section>

```

```

    </div>
  </section> </div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Customer profile page(profil.php)

```

<?php
session_start();
if (!isset($_SESSION['email'])) { header("Location: index.php"); exit; }
include('includes/config.php');
?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1.0">
<title>My Profile - Digital Dairy</title>
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<!-- HERO -->
<section class="text-center py-5" style="background:linear-gradient(135deg,#f8f9fa,#dee2e6);">
  <div>
    <div style="width:100px;height:100px;background:linear-gradient(135deg,#2c5aa0,#4caf50);border-radius:50%;display:flex;align-items:center;justify-content:center;margin:auto;">
      <i class="fas fa-user-circle text-white fs-1"></i>
    </div>
    <h1 class="fw-bold mt-3" style="background:linear-gradient(135deg,#2c5aa0,#4caf50);-webkit-background-clip:text;-webkit-text-fill-color:transparent;">My Profile</h1>
    <p class="text-secondary">Manage your account and preferences</p>
  </div>
</section>
<section class="py-5">
  <div class="container">
    <div class="card p-4 shadow-lg border-0 rounded-4">

```

```

<h3 class="text-primary mb-4"><i class="fas fa-id-card me-2"></i>Profile Information</h3>
<div class="row g-3">
  <div class="col-md-6"><strong>Name:</strong> <?php echo htmlspecialchars($user_name); ?></div>
  <div class="col-md-6"><strong>Email:</strong> <?php echo htmlspecialchars($user_email); ?></div>
  <div class="col-md-6"><strong>Phone:</strong> <?php echo htmlspecialchars($user_data['contact']) ?? 'Not provided'; ?></div>
  <div class="col-md-6"><strong>Member Since:</strong> <?php echo date('F Y', strtotime($user_data['registration_date'] ?? 'now')); ?></div>
</div>
<hr class="my-4">
<h4 class="text-primary mb-3"><i class="fas fa-edit me-2"></i>Edit Profile</h4>
<form method="POST" action="profile.php">
  <div class="row g-3">
    <div class="col-md-6"><input type="text" name="customer_name" class="form-control" value="<?php echo htmlspecialchars($user_name); ?>" required></div>
    <div class="col-md-6"><input type="tel" name="phone" class="form-control" value="<?php echo htmlspecialchars($user_data['contact']) ?? ''; ?>"></div>
    <div class="col-12"><textarea name="address" class="form-control" rows="3"><?php echo htmlspecialchars($user_data['address']) ?? ''; ?></div>
  </div>
  <div class="mt-4 d-flex gap-2">
    <button type="submit" name="update_profile" class="btn btn-success"><i class="fas fa-save me-1"></i>Update</button>
    <a href="home_page.php" class="btn btn-outline-secondary"><i class="fas fa-arrow-left me-1"></i>Back</a>
  </div>
</form> </div> </div></section>
<!-- STATS -->
<section class="py-5 bg-light">
  <div class="container">
    <div class="card p-4 shadow border-0 rounded-4 text-center">
      <h4 class="text-primary mb-4"><i class="fas fa-chart-bar me-2"></i>Account Stats</h4>
      <div class="row">
        <div class="col-md-4 mb-3">
          <div class="p-3 bg-primary-subtle rounded"><i class="fas fa-shopping-bag fs-3 text-primary"></i><br>
          Orders:
          <?php
            $r = mysqli_query($con, "SELECT COUNT(*) c FROM tblorders WHERE Email='$user_email'");
            echo mysqli_fetch_assoc($r)['c'];

```

```

        ?> </div> </div>

<div class="col-md-4 mb-3">

    <div class="p-3 bg-success-subtle rounded"><i class="fas fa-rupee-sign fs-3 text-success"></i><br>

        ₹<?php

            $r = mysqli_query($con, "SELECT SUM(TotalAmount) t FROM tblorders WHERE
Email='$user_email'");

            echo number_format(mysqli_fetch_assoc($r)['t'] ?? 0, 2);

        ?>

    </div> </div>

<div class="col-md-4 mb-3">

    <div class="p-3 bg-warning-subtle rounded"><i class="fas fa-calendar fs-3 text-warning"></i><br>

        <?php echo date('M Y', strtotime($user_data['registration_date'] ?? 'now')); ?>

    </div> </div> </div> </div> </div> </section>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>

</body>

</html>

```

Product order page(orders.php)

```

<?php
session_start();

if (!isset($_SESSION['email'])) { header("Location: index.php"); exit; }

include('includes/config.php');

if (!isset($_SESSION['cart'])) $_SESSION['cart'] = [];

$user_email = $_SESSION['email'];

$cart_count = count($_SESSION['cart']);

// Fetch orders

$q = "SELECT * FROM tblorders WHERE Email='$user_email' ORDER BY OrderDate DESC";

$r = mysqli_query($con, $q);

$total_orders = $pending_orders = $delivered_orders = 0;

if ($r && mysqli_num_rows($r) > 0) {

    $total_orders = mysqli_num_rows($r);

    while ($o = mysqli_fetch_assoc($r)) {

        if ($o['Status'] == 'Pending') $pending_orders++;

        if ($o['Status'] == 'Delivered') $delivered_orders++;

    }

}

```

```

        mysqli_data_seek($r, 0);
    }
    if (isset($_POST['action'])) {
        $id = $_POST['order_id'] ?? "";
        if ($_POST['action'] == 'view_details') {
            header("Location: order_details.php?id=$id");
            exit;
        } elseif ($_POST['action'] == 'reorder') {
            $items = mysqli_query($con, "SELECT * FROM tblorderitems WHERE OrderID='$id'");
            if ($items && mysqli_num_rows($items) > 0) {
                while ($i = mysqli_fetch_assoc($items)) {
                    $pid = $i['ProductID']; $qty = $i['Quantity'];
                    if (isset($_SESSION['cart'][$pid])) {
                        $_SESSION['cart'][$pid]['quantity'] += $qty;
                    } else {
                        $p = mysqli_query($con, "SELECT * FROM tblproduct WHERE ID='$pid'");
                        if ($p && mysqli_num_rows($p) > 0) {
                            $prod = mysqli_fetch_assoc($p);
                            $_SESSION['cart'][$pid] = [
                                'id' => $prod['ID'], 'name' => $prod['ProductName'],
                                'price' => $prod['UnitPrice'], 'image' => $prod['ProductImage'],
                                'quantity' => $qty
                            ];
                        }
                    }
                }
                header("Location: cart.php"); exit;
            }
        }
    }
}

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1.0">
<title>My Orders - Digital Dairy</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.0.0/css/all.min.css" rel="stylesheet">
</head>
<body>

```

```

<!-- NAVBAR -->
<nav class="navbar navbar-expand-lg bg-white shadow-sm fixed-top">
  <div class="container">
    <a class="navbar-brand fw-bold text-primary" href="home_page.php">DDMS</a>
    <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#nav"><span class="navbar-
toggler-icon"></span></button>
    <div class="collapse navbar-collapse" id="nav">
      <ul class="navbar-nav me-auto">
        <li class="nav-item"><a href="home_page.php" class="nav-link">Home</a></li>
        <li class="nav-item"><a href="home_page.php#about" class="nav-link">About</a></li>
        <li class="nav-item"><a href="home_page.php#contact" class="nav-link">Contact</a></li>
      </ul>
      <a href="cart.php" class="btn btn-outline-primary me-3 position-relative">
        <i class="fas fa-shopping-cart"></i>
        <?php if ($cart_count > 0): ?>
          <span class="badge bg-danger position-absolute top-0 start-100 translate-middle"><?php echo
$cart_count; ?></span>
        <?php endif; ?>
      </a>
      <div class="dropdown">
        <button class="btn btn-outline-primary dropdown-toggle" data-bs-toggle="dropdown">
          <i class="fas fa-user"></i> <?php echo htmlspecialchars($_SESSION['customer_name'] ?? 'User'); ?>
        </button>
        <ul class="dropdown-menu dropdown-menu-end">
          <li><a class="dropdown-item" href="profile.php">My Profile</a></li>
          <li><a class="dropdown-item active" href="orders.php">My Orders</a></li>
          <li><hr class="dropdown-divider"></li>
          <li><a class="dropdown-item" href="logout.php">Logout</a></li>
        </ul>
      </div>
    </div>
  </nav>
<div style="height:80px"></div>
<!-- HERO -->
<section class="py-5 text-center" style="background:linear-gradient(135deg,#f8f9fa,#dee2e6);">
  <div class="container">
    <h1 class="fw-bold" style="background:linear-gradient(135deg,#2c5aa0,#4caf50);-webkit-background-
clip:text;-webkit-text-fill-color:transparent;">My Orders</h1>

```



```

        <p class="text-muted mb-4">Track your order history and manage purchases</p>
        <a href="home_page.php" class="btn btn-outline-primary me-2"><i class="fas fa-arrow-left me-1"></i>Shop
        More</a>

        <a href="cart.php" class="btn btn-primary"><i class="fas fa-credit-card me-1"></i>Checkout</a>
    </div>
</section>
<!-- SUMMARY -->
<section class="py-4 bg-light">
    <div class="container d-flex justify-content-center">
        <div class="p-3 bg-white rounded shadow-sm" style="min-width:280px">
            <h5 class="text-center text-primary mb-3">Order Summary</h5>
            <div class="d-flex justify-content-between"><span>Total:</span><b><?php echo $total_orders;
            ?></b></div>
            <div class="d-flex justify-content-between text-warning"><span>Pending:</span><b><?php echo
            $pending_orders; ?></b></div>
            <div class="d-flex justify-content-between text-success"><span>Delivered:</span><b><?php echo
            $delivered_orders; ?></b></div>
        </div>
    </div>
</section>
<!-- ORDERS -->
<section class="py-5">
    <div class="container">
        <?php if (!$r || mysqli_num_rows($r) == 0): ?>
            <div class="text-center py-5">
                <i class="fas fa-shopping-bag fa-4x text-secondary mb-3"></i>
                <h3>No orders found</h3>
                <p class="text-muted">You haven't placed any orders yet.</p>
                <a href="home_page.php" class="btn btn-primary">Start Shopping</a>
            </div>
        <?php else: ?>
            <h3 class="text-primary mb-4">Order History</h3>
            <?php while ($o = mysqli_fetch_assoc($r)):
                $it = mysqli_query($con, "SELECT oi.*,p.ProductName,p.ProductImage FROM tblorderitems oi JOIN
                tblproduct p ON oi.ProductID=p.ID WHERE oi.OrderID='".$o['ID']."'");
                $count = mysqli_num_rows($it);
            ?>

```

```

<div class="p-4 mb-4 bg-white shadow-sm rounded" style="border-left:5px solid <?php echo
$so['Status']== 'Delivered'? '#4caf50': '#ff6b35'; ?>">

    <div class="d-flex justify-content-between align-items-center mb-2">

        <h5 class="text-primary mb-0">Order #<?php echo $so['ID']; ?></h5>

        <span class="badge text-white px-3 py-2" style="background:<?php echo
$so['Status']== 'Delivered'? '#4caf50': ($so['Status']== 'Pending'? '#ff6b35': '#2c5aa0'); ?>">

            <i class="fas fa-<?php echo $so['Status']== 'Delivered'? 'check-
circle': ($so['Status']== 'Pending'? 'clock': 'truck'); ?>"></i> <?php echo $so['Status']; ?>

        </span>

    </div>

    <small class="text-muted"><i class="fas fa-calendar"></i> <?php echo date('M d, Y h:i A',
strtotime($so['OrderDate'])); ?></small>

</div>

<div class="mt-3 d-flex align-items-center">

    <?php if ($it && mysqli_num_rows($it) > 0): $i = 0;

        while ($p = mysqli_fetch_assoc($it)) {

            if ($i++ < 3) echo '';

        }

        if ($count > 3) echo '<div class="rounded bg-light px-3 py-2">+'.($count-3).</div>';

    endif; ?>

    <div class="ms-3">

        <b><?php echo $count; ?> item<?php echo $count>1?'s':"; ?></b><br>

        <small>Total: ₹<?php echo number_format($so['TotalAmount'], 2); ?></small>

    </div>

</div>

<div class="mt-3">

    <form method="POST" class="d-inline">

        <input type="hidden" name="action" value="view_details">

        <input type="hidden" name="order_id" value="<?php echo $so['ID']; ?>">

        <button class="btn btn-outline-primary btn-sm"><i class="fas fa-eye"></i> View</button>

    </form>

    <?php if ($so['Status']== 'Delivered'): ?>

    <form method="POST" class="d-inline">

        <input type="hidden" name="action" value="reorder">

        <input type="hidden" name="order_id" value="<?php echo $so['ID']; ?>">

        <button class="btn btn-outline-success btn-sm"><i class="fas fa-redo"></i> Reorder</button>

    </form>

```

```

        <?php endif; ?>    </div>    </div>

    <?php endwhile; ?>

    <?php endif; ?>

</div>

</section>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>

</body>

</html>

```

Cart page(cart.php)

```

<?php
session_start();

if (!isset($_SESSION['email'])) { header("Location: index.php"); exit; }

include('includes/config.php');

$cart = $_SESSION['cart'] ?? [];

$subtotal = array_sum(array_map(fn($i) => $i['price'] * $i['quantity'], $cart));

$total = $subtotal;

?>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1">

<title>Shopping Cart - DDMS</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">

<link href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.4.0/css/all.min.css" rel="stylesheet">

</head>

<body style="font-family:Poppins,sans-serif;">

<div style="height:80px"></div>

<section class="py-5 bg-light text-center">

    <h1 class="fw-bold" style="color:#2c5aa0;">Shopping Cart</h1>

    <p class="text-muted">Review your selected items and proceed to checkout</p>

    <a href="home_page.php" class="btn btn-outline-primary"><i class="fa fa-arrow-left me-2"></i>Continue Shopping</a>

</section>

<div class="container py-5">

```

```

<?php if(empty($cart)): ?>
    <div class="text-center py-5">
        <i class="fa fa-shopping-cart fa-4x text-muted mb-3"></i>
        <h3 class="text-primary">Your cart is empty</h3>
        <a href="home_page.php" class="btn btn-primary mt-3">Start Shopping</a>
    </div>
<?php else: ?>
    <div class="row">
        <div class="col-lg-8">
            <?php foreach($cart as $id=>$item): ?>
                <div class="d-flex align-items-center justify-content-between border rounded p-3 mb-3 shadow-sm">
                    <div class="d-flex align-items-center">
                        
                        <div>
                            <h6 class="mb-1 text-primary"><?=$item['name']?></h6>
                            <small><?=>₹<?=number_format($item['price'],2)?> × <?=$item['quantity']?></small>
                        </div>
                    </div>
                    <div class="text-end">
                        <strong><?=>₹<?=number_format($item['price']*$item['quantity'],2)?></strong><br>
                        <form method="POST" style="display:inline;">
                            <input type="hidden" name="action" value="remove">
                            <input type="hidden" name="product_id" value="<?=$id?>">
                            <button class="btn btn-sm btn-outline-danger mt-2"><i class="fa fa-trash"></i></button>
                        </form>
                    </div>
                </div>
            <?php endforeach; ?>
        </div>
        <div class="col-lg-4">
            <div class="p-4 rounded shadow-sm bg-white sticky-top">
                <h5 class="text-primary mb-3">Order Summary</h5>
                <div
                    class="d-flex
                    justify-content-between">
                    <span>Subtotal:</span>
                    <strong><?=>₹<?=number_format($subtotal,2)?></strong>
                </div>
                <hr>
                <div
                    class="d-flex
                    justify-content-between">
                    <span>Total:</span>
                    <strong><?=>₹<?=number_format($total,2)?></strong>
                </div>
                <form method="POST" action="process_checkout.php" class="mt-3">

```

```

                <button type="submit" class="btn btn-success w-100"><i class="fa fa-credit-card me-2"></i>Checkout</button>
            </form>
        </div> </div> </div>
<?php endif; ?>
</div></body></html>

```

Logout page(logout.php)

```

<?php
session_start();
session_unset();
session_destroy();
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Logging Out...</title>
    <script>
        // Redirect to login page after 2 seconds
        setTimeout(function() {
            window.location.href = "login.php";
        }, 2000);
    </script>
</head>
<body>
    <div class="logout-box">
        <h2>✅ Logged Out Successfully</h2>
        <p>Redirecting to login page...</p>
    </div>
</body>
</html>

```

5.2.2 CODE VALIDATION AND OPTIMIZATION

Code validation ensures that the written code meets syntax, logic, and security standards. Optimization enhances performance and maintainability. The following practices were applied in the DDMS project:

a) Input Validation

- All user inputs are validated on both client-side (using JavaScript) and server-side (using PHP).
- Mandatory fields are checked to avoid incomplete form submission.
- File uploads are restricted to specific formats (e.g., .jpg, .png).

b) Database Optimization

- Indexing was used on frequently queried columns such as farmer_uuid, product_type, and sales_date.
- Queries were structured using **MySQLi** for faster execution and security.
- Redundant queries were minimized by creating reusable database functions.

c) Security Enhancements

- Implemented **session-based authentication** to restrict unauthorized access.
- Used **MD5 hashing** for passwords (though future versions will use password_hash() for stronger encryption).
- Prevented **SQL injection** using input sanitization and escape functions.
- Validated file uploads to avoid malicious scripts.

d) Code Reusability and Maintainability

- Common layout files (header, footer, sidebar) were included via PHP include() statements.
- Repeated database queries were modularized.
- Organized folder structure for easy navigation and updates.

e) UI Optimization

- Used **Bootstrap grid system** for responsive layouts.
- Included **DataTables** for managing and filtering large data efficiently.
- Applied **Chart.js** for smooth rendering of graphs and reports.

5.3 UNIT TESTING

Unit testing is the process of testing individual components of the software to ensure each performs correctly in isolation. The objective is to identify and fix bugs early in development.

5.3.1 Testing Approach

Each module in the DDMS was tested independently using manual testing techniques. Test cases were created for various functionalities, including:

- Form submission validation
- Database operations (Insert, Update, Delete, Select)
- Login and logout flow
- Session handling
- Data filtering and report generation

All errors found during testing were corrected immediately, ensuring smooth functionality across modules.

5.3.2 Sample Unit Test Results

Module	Test Description	Expected Result	Actual Result	Status
Admin Login	Valid and invalid credentials	Access granted / denied appropriately	As expected	Passed
Add Farmer	Add new farmer record	Farmer details saved in farmers table	As expected	Passed
Milk Collection	Record daily milk entry	Data inserted with correct date and farmer UUID	As expected	Passed
Add Product	Add product with image upload	Product stored with image path	As expected	Passed
Delivery Module	Add and manage deliveries	Delivery details saved and displayed	As expected	Passed
Sales Report	Generate report for given date range	Accurate total and invoice displayed	As expected	Passed
Customer Order Flow	Place order, checkout, and confirm	Order saved in tblorders and displayed	As expected	Passed

SYSTEM TESTING

CHAPTER 6

SYSTEM TESTING

6.1 INTRODUCTION

System testing is an essential phase in the software development life cycle where the complete and integrated system is evaluated against its defined requirements. It ensures that the developed application performs its intended functions correctly, efficiently, and reliably under real-world operating conditions.

For the Digital Dairy Management System (DDMS), testing was performed on all functional modules, including Farmer Management, Milk Collection, Product Inventory, Sales, Delivery, and Reports.

The main objectives of testing were to:

- Verify that the system meets all functional and non-functional requirements.
- Ensure the correct flow of data between modules.
- Identify and correct any defects prior to deployment.
- Validate that security, usability, and performance standards are met.

The testing process followed a structured approach consisting of **unit testing**, **integration testing**, and **system testing**—in alignment with the **Waterfall model** used for development.

6.2 INTEGRATION TESTING

Integration testing focuses on verifying the data flow and interactions between interconnected modules. Once each module was individually tested and validated through unit testing, they were combined and tested as an integrated system.

6.2.1 Objective

The objective of integration testing in DDMS was to ensure that:

- Data passed correctly between modules (e.g., Farmer → Milk Collection → Payment).
- The admin panel could access, modify, and report data from multiple modules simultaneously.
- No data loss, duplication, or inconsistency occurred when integrating modules.

6.2.2 Integration Testing Approach

An **incremental integration testing** strategy was adopted. Modules were integrated and tested one by one to simplify debugging and ensure smooth flow of control:

1. **Admin Login and Database** → Tested authentication and database connectivity.
2. **Farmer + Milk Collection** → Validated daily milk collection entries linked to farmer profiles.
3. **Product + Sales** → Ensured product stock updated automatically after each sale.
4. **Sales + Delivery** → Verified that order status updates reflected correctly in the sales module.
5. **Dashboard + Reports** → Confirmed accurate real-time analytics using Chart.js and DataTables.

6.2.3 Integration Testing Results

Modules Integrated	Test Description	Expected Result	Actual Result	Status
Admin + Database	Verify admin login and DB connection	Successful login, access granted	Works correctly	Passed
Farmer + Milk Collection	Check linkage between farmer records and milk entries	Records linked to correct farmer ID	Works correctly	Passed
Product + Sales	Verify stock updates after sale entry	Quantity reduced in inventory	Works correctly	Passed
Sales + Delivery	Ensure delivery status syncs with sales	Delivery marked “Completed” in report	Works correctly	Passed
Dashboard + Reports	Verify data accuracy in analytics	Reports display correct totals and charts	Works correctly	Passed

All integrated modules functioned as expected without data loss or inconsistencies.

6.3 SYSTEM TESTING

System testing validates the entire software product as a unified whole. The goal is to confirm that the **Digital Dairy Management System** operates according to the requirements specified during analysis and design phases.

6.3.1 Testing Objectives

- Verify that all modules work together as expected.
- Ensure proper role-based access control (Admin, Farmer, Customer).
- Confirm that CRUD (Create, Read, Update, Delete) operations perform correctly.

- Validate security mechanisms like session management and password hashing.
- Test responsiveness, data accuracy, and user interface consistency.

6.3.2 Testing Environment

- **Operating System:** Windows 10 / Linux Ubuntu
- **Web Server:** Apache (via XAMPP)
- **Database:** MySQL 8.0
- **Browser:** Google Chrome, Microsoft Edge
- **Language:** PHP 8, HTML5, CSS3, JavaScript
- **Frameworks:** Bootstrap 5, Chart.js, DataTables

6.4 Test Plan & Test Case

Test Area	Objective	Testing Type	Expected Output
Login System	Validate admin and user authentication	Functional Testing	Correct access or error message
Farmer Module	Verify CRUD operations for farmers	Functional Testing	Data saved and displayed accurately
Milk Collection	Ensure daily records are linked to farmer ID	Integration Testing	Accurate quantity and payment records
Product Module	Manage inventory with image upload	Functional Testing	Product added/edited successfully
Sales Module	Validate invoice generation	System Testing	Correct totals and calculations
Delivery Module	Track deliveries and statuses	System Testing	Real-time delivery updates
Report Module	Generate analytics and exports	Functional Testing	Correct filtered reports
Security	Check session handling and password protection	Security Testing	Secure access and logout behavior

Below are representative test cases from major DDMS modules:

Test Case ID	Module	Test Description	Input / Action	Expected Output	Actual Output	Status
TC-01	Login	Admin login with valid credentials	admin@gmail.com / admin	Redirect to dashboard	Works as expected	Passed
TC-02	Login	Invalid login credentials	admin@gmail.com / wrong	Show “Invalid login” alert	Works as expected	Passed
TC-03	Farmer	Add new farmer details	Form inputs + Save	Farmer saved in database	Works as expected	Passed
TC-04	Milk Collection	Record daily milk data	Farmer ID + Quantity	Record saved successfully	Works as expected	Passed
TC-05	Product	Add new product	Name + Price + Image	Product displayed in list	Works as expected	Passed
TC-06	Sales	Generate new invoice	Select customer + product	Invoice created with total amount	Works as expected	Passed
TC-07	Delivery	Update delivery status	Change to “Delivered”	Status updated in DB	Works as expected	Passed
TC-08	Reports	Generate report by date	Select date range	Data filtered correctly	Works as expected	Passed
TC-09	Logout	End session	Click logout button	Redirect to login page	Works as expected	Passed

SYSTEM MAINTENANCE

CHAPTER 7

SYSTEM MAINTENANCE

7.1 Introduction

After successful implementation and testing, the next important stage is System Evaluation and Maintenance. This phase ensures that the **Digital Dairy Management System (DDMS)** performs efficiently in real-world conditions and continues to meet the needs of administrators, farmers, and customers. System evaluation determines whether the developed system satisfies user requirements, performs as expected, and improves the overall workflow of dairy management. Maintenance focuses on identifying and resolving issues that may arise during daily operation, ensuring the system's long-term reliability and scalability.

7.2 System Maintenance

Maintenance is an ongoing process aimed at improving system performance, fixing errors, and adding new features as the organization's needs evolve. The Digital Dairy Management System follows three main types of maintenance activities:

7.2.1 Corrective Maintenance

This involves fixing bugs or errors that are identified during regular system usage. For example:

- Resolving form validation issues.
- Correcting incorrect data displays in reports.
- Fixing SQL or PHP syntax errors discovered after deployment.

7.2.2 Adaptive Maintenance

As dairy operations expand, new requirements may arise. Adaptive maintenance ensures that the system evolves accordingly.

Examples include:

- Adding new modules (like “Customer Portal” or “Online Order Tracking”).
- Updating PHP or MySQL versions to maintain compatibility.
- Enhancing features such as automatic report generation.

7.2.3 Perfective Maintenance

These are improvements made to enhance performance and usability beyond the original design.

Examples include:

- Optimizing queries for faster data retrieval.
- Improving dashboard analytics and charts.
- Enhancing the visual UI for better user experience.

FUTURE ENHANCEMENT AND SCOPE OF FURTHER DEVELOPMENT

CHAPTER 8

FUTURE ENHANCEMENT AND SCOPE OF FURTHER DEVELOPMENT

8.1 INTRODUCTION

The Digital Dairy Management System (DDMS) is a web-based solution developed using PHP, MySQL, HTML, CSS, and Bootstrap to automate dairy operations such as milk collection, product management, sales tracking, delivery management, and reporting. It has greatly improved the efficiency and accuracy of managing daily dairy activities for farmers, administrators, and customers. However, as with any software, there is always room for improvement. This chapter discusses the **key merits, limitations, and future enhancements** that can be implemented to make the system more efficient, scalable, and user-friendly in the future.

8.2 MERITS OF THE SYSTEM

1. **Automation of Dairy Operations**
The system automates routine tasks like milk collection entry, product management, and sales tracking, minimizing manual work.
2. **Centralized Database**
All farmer, product, and sales information is stored in a single database, ensuring consistency and easy data retrieval.
3. **User-Friendly Interface**
The interface is built with Bootstrap and is responsive, clean, and simple to use for users with limited technical knowledge.
4. **Improved Accuracy**
Automated calculations and digital record-keeping eliminate human errors and improve transaction accuracy.
5. **Role-Based Access**
The system ensures data security by assigning different access rights for admin, farmers, and customers.
6. **Real-Time Reports**
Sales, milk collection, and product reports can be generated quickly for better decision-making.
7. **Cost-Effective Solution**
Since it uses open-source technologies, it requires minimal maintenance cost and can be deployed easily.

8.3 LIMITATIONS OF THE SYSTEM

1. Limited User Roles

The system currently provides access mainly for admin and farmers, without a separate panel for customers or delivery staff.

2. Lack of Mobile Application

The system is web-based and optimized for browsers but lacks dedicated Android or iOS applications.

3. No Online Payment Integration

Transactions are recorded manually; there is no direct payment gateway for digital payments.

4. No Automated Notifications

The system does not send real-time SMS or email notifications for order updates, payments, or milk collection alerts.

5. Basic Analytics

Reports are static and basic; there are no data visualizations or advanced analytics.

6. Manual Backup System

Data backups must be done manually, which may risk data loss if not performed regularly.

8.4 FUTURE ENHANCEMENT OF THE SYSTEM

1. Mobile Application Development

Develop Android and iOS apps for farmers, customers, and administrators to access and manage data on the go.

2. Integration of Payment Gateway

Include secure online payment systems such as Razorpay, Paytm, or UPI for seamless transactions.

3. Customer and Delivery Staff Modules

Create dedicated dashboards for customers to track orders and for delivery staff to manage and update delivery statuses.

4. Real-Time Notifications

Implement SMS and email notifications for order confirmation, milk collection updates, and payment reminders.

5. Advanced Data Analytics

Add graphical reports, trend analysis, and AI-based predictive analytics for better decision-making.

6. Automated Backup and Cloud Integration

Enable cloud-based or automated database backup systems for data safety and easy recovery.

7. Multi-Language Support

Add regional languages like Malayalam, Hindi, or Tamil to make the system accessible to local farmers.

8. Barcode and RFID Integration

Implement RFID/barcode technology for tracking milk cans, inventory, and deliveries in large-scale dairies.

9. Enhanced Security Mechanisms

Upgrade password encryption, implement two-factor authentication, and maintain detailed activity logs for data security.

10. IoT Integration (Long-Term Goal)

Integrate IoT sensors for automatic milk quality monitoring (fat content, temperature) directly from the collection point.

APPENDIX, CONCLUSION & BIBLIOGRAPHY

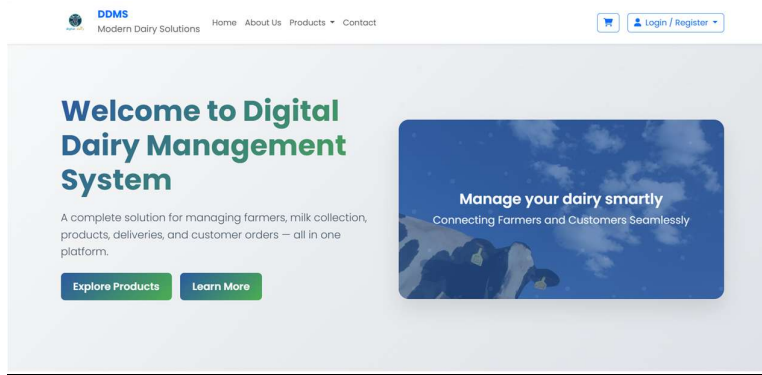
CHAPTER 9

APPENDIX, CONCLUSION & BIBLIOGRAPHY

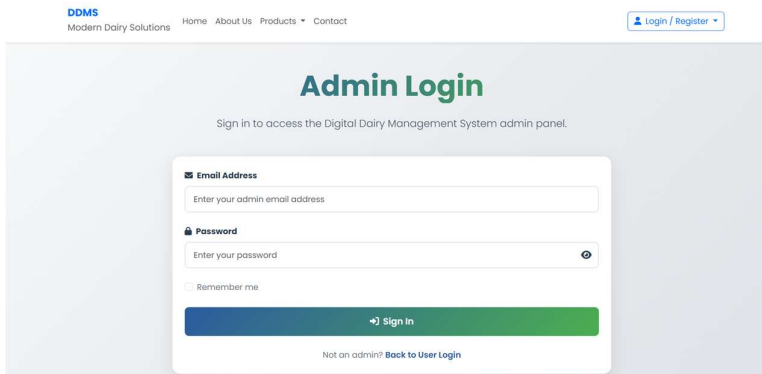
9.1 Appendix

Screenshots

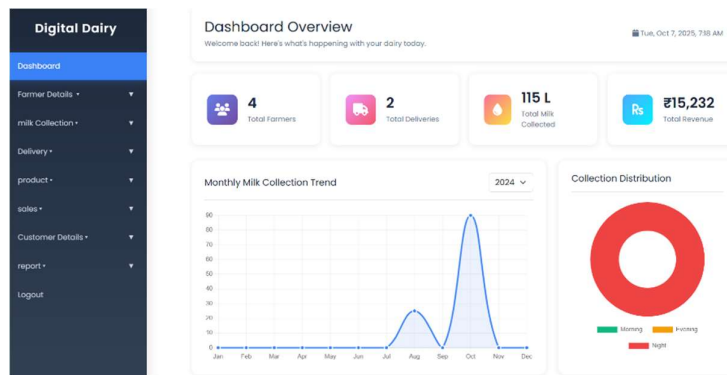
Home page



Admin pages



Admin view-login page



Admin view- The main administrative dashboard providing a comprehensive overview of the dairy management system

Digital Dairy

Add New Farmer
Register a new farmer in the dairy management system

Farmer Registration Form
Please fill in all required fields to register a new farmer

Fields:

- Full Name ***: Enter farmer's full name
- Username ***: Enter unique username. Choose a unique username for easy identification.
- Contact Number**: Enter 10-digit phone number. Enter a valid 10-digit phone number (e.g., 9876543210).
- Address ***: Enter complete address
- Farm Size**: 0.0 acres
- Email Address ***: Enter email address
- Aadhar Number**: Enter 12-digit Aadhar number

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Admin view- Farmer registration form for adding new farmers to the system

Digital Dairy

Manage Farmers
View, edit, and manage all registered farmers in the system

Farmer Management
Manage all farmer records and their details

Add New Farmer

All Farmers
Comprehensive list of all registered farmers

Show 10 farmers per page

Search farmers:

# ID	Name	Username	Status	Contact	Address	Farm Size	Email	Registered	Actions
LL40	pooja Aadhar: 123456789012	@poor021	Active	9847012251	konnai...	1.50 acres	poorja@gmail.com	Oct 4, 2025	[Edit] [Delete] [Add]
HVFU	shibin Aadhar: 123456789009	@shibin01	Active	9847012251	ssssssss...	0.20 acres	shibin@gmail.com	Oct 4, 2025	[Edit] [Delete] [Add]

Admin view- Farmer management dashboard

Back to go back, hold to see history

Digital Dairy

Dashboard

Farmer Details ▾

Milk Collection ▾

Delivery ▾

Product ▾

Sales ▾

Customer Details ▾

Report ▾

Logout

Add Milk Collection Wed, Oct 8, 2025, 3:44 PM

Record new milk collection data from farmers

Milk Collection Entry

Enter milk collection details from registered farmers

[View All Records](#)

Milk Collection Form Please fill in all required fields to record milk collection

Date * 08-10-2025

Select Farmer * -- Select Farmer --

Farmer UUID

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Admin view- A form page for recording milk collection data from farmers

Digital Dairy

Dashboard

Farmer Details ▾

Milk Collection ▾

Delivery ▾

Product ▾

Sales ▾

Customer Details ▾

Report ▾

Logout

Manage Milk Collection Wed, Oct 8, 2025, 3:45 PM

View and manage all milk collection records

Milk Collection Records

Comprehensive list of all milk collection entries

[Add New Collection](#)

All Milk Collections View and search through all milk collection records

# ID	Date	Farmer	Product Type	Quantity	Fat Content	Temperature	Payment
1	Oct 4, 2025	Joyal royce	Milk	56.80L	4.56%	8.0°C	₹ 666.00
2	Oct 4, 2025	Pooja	Milk	54.80L	4.98%	4.0°C	₹ 45.00

Total Volume: 115.00 L

Total Revenue: ₹4,099.00

Total Records: 4

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Admin view- Comprehensive milk collection management dashboard

Digital Dairy

Dashboard

Farmer Details ▾

Milk Collection ▾

Delivery ▾

Product ▾

Sales ▾

Customer Details ▾

Report ▾

Logout

Add Delivery Details Wed, Oct 8, 2025, 3:45 PM

Schedule a new delivery for customers

Delivery Scheduling

Schedule new delivery orders for customers

[View All Deliveries](#)

Delivery Form Please fill in all required fields to schedule a delivery

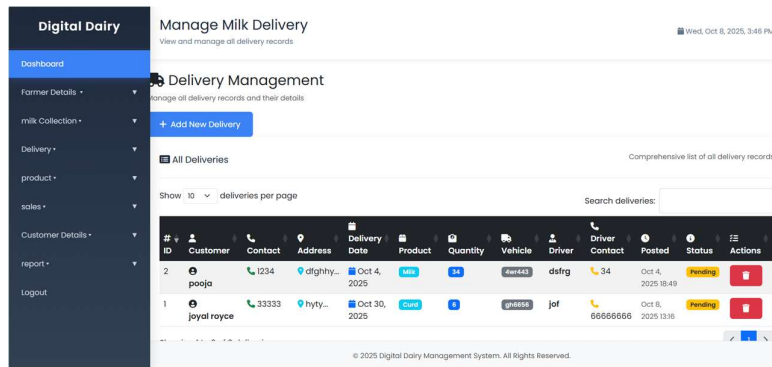
Customer Name * Enter customer name

Customer Contact * Enter phone number

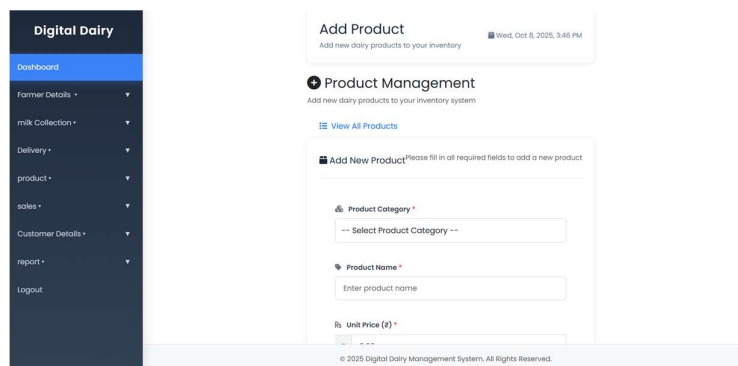
Delivery Address * Enter delivery address

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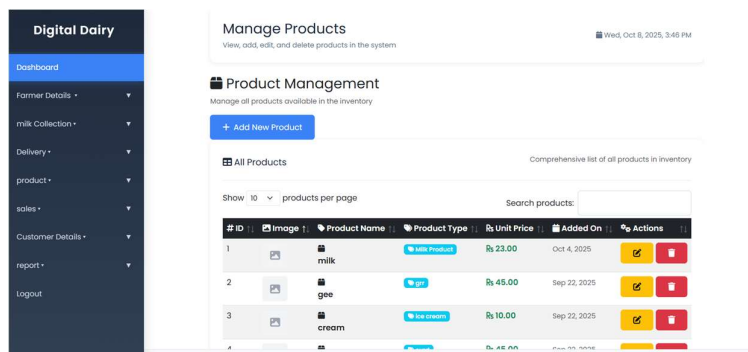
Admin view- A delivery scheduling interface for creating new delivery orders



Admin view- Delivery management and tracking system.



Admin view- Product management form for adding new dairy products to inventory



Admin view- Product inventory management system.

Refresh Data

Digital Dairy

Dashboard

Farmer Details

milk Collection

Delivery

product

sales

Customer Details

report

Logout

Add Sales Record

Record new sales transactions

Wed, Oct 8, 2025, 3:49 PM

Sales Management

Record new sales transactions and manage customer orders

View All Sales

Sales Entry Form

Please fill in all required fields to record a new sale

Invoice Number

e.g., INV-2024-001

Customer Name

Enter customer name

Contact

Enter contact number

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Admin view- Sales transaction recording interface

Digital Dairy

Dashboard

Farmer Details

milk Collection

Delivery

product

sales

Customer Details

report

Logout

Sales Management

Manage all sales records and customer transactions

+ Add New Sale

View Reports

All Sales Records

Comprehensive list of all sales transactions

Show 10 sales per page

Search sales:

#	ID	Invoice	Customer	Contact	Product	Quantity	Price	Total	Date	Actions
1	3434		pooja	23445	milk	34.00 units	4.00	Rs 136.00	Oct 4, 2025	View Edit Delete
2	qw23233		pooja	1234567	milk	34.00 units	444.00	Rs 15,096.00	Oct 4, 2025	View Edit Delete

Showing 1 to 2 of 2 sales

2 Total Sales

₹15,232.00 Total Revenue

0 Today's Sales

2 This Month

Admin view- Sales management and analytics dashboard.

Digital Dairy

Dashboard

Farmer Details

milk Collection

Delivery

product

sales

Customer Details

report

Logout

Manage Customers

View and manage all registered customers

Wed, Oct 8, 2025, 3:49 PM

Customer Management

Manage customer accounts and information

+ Add New Customer

Customer Report

2 Total Customers

2 Active Customers

0 Today's Registrations

Customer List

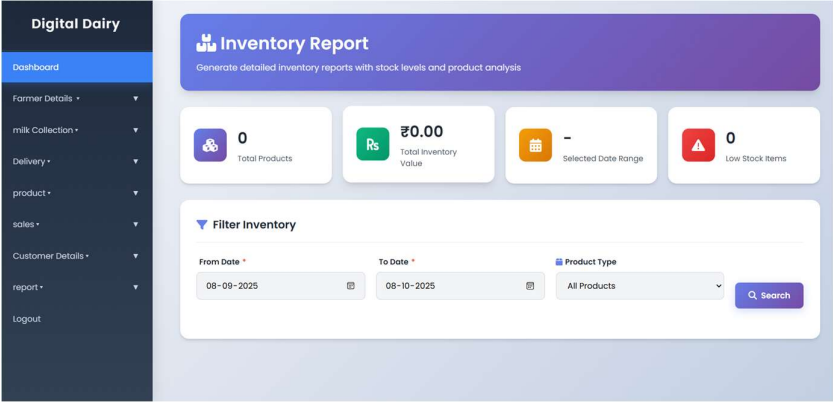
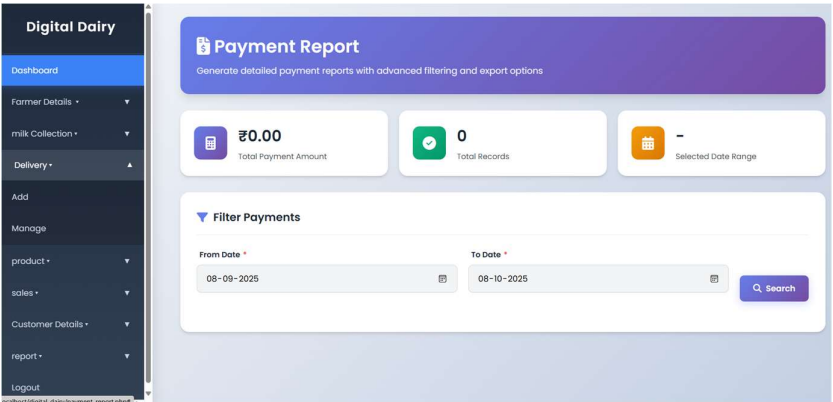
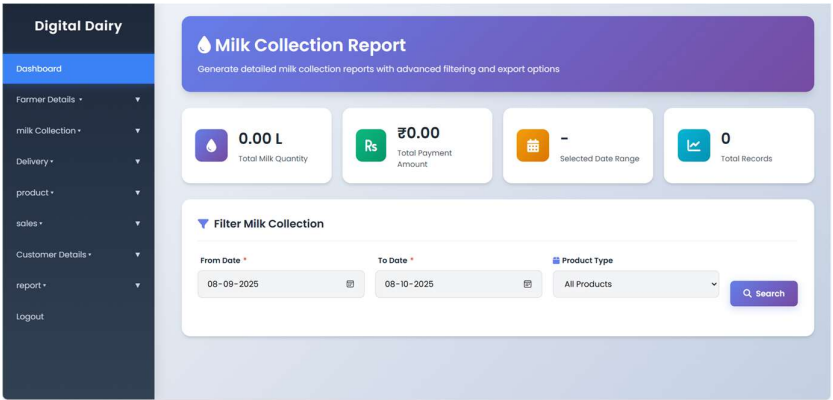
All registered customers in the system

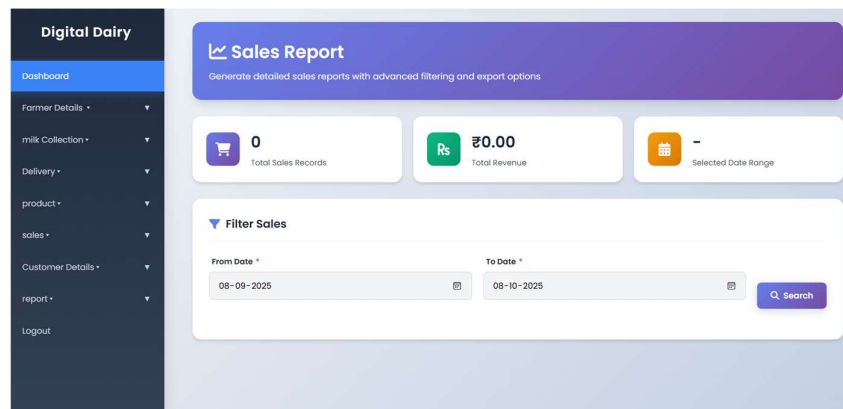
#	ID	Name	Contact	Email	Address	Reg. Date	Status	Actions
8		jas		jas@gmail.com		04 Oct 2025	Active	View Edit Delete
6		joyal		joyal@gmail.com	alunkal house mallappally nart...	04 Oct 2025	Active	View Edit Delete

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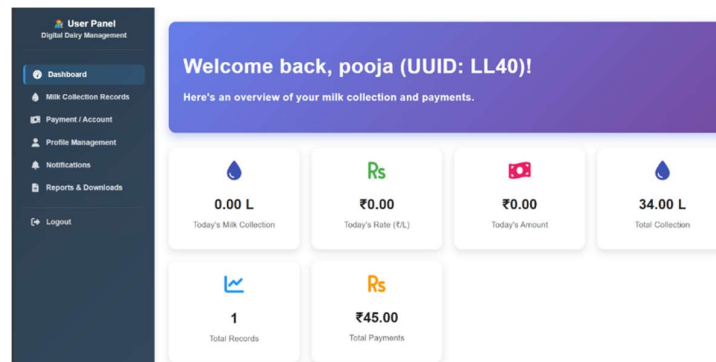
Admin view- Customer data management interface

Report pages:

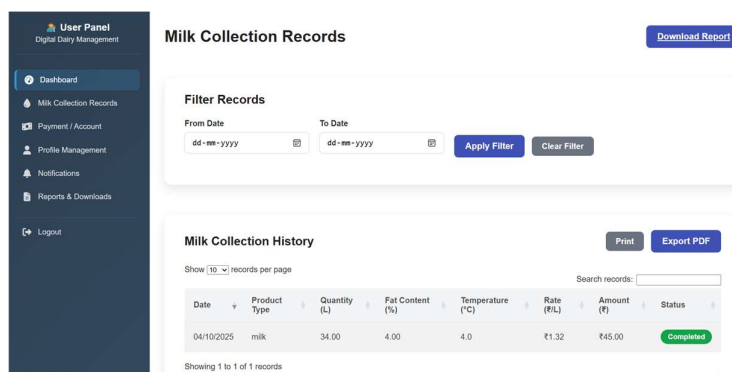




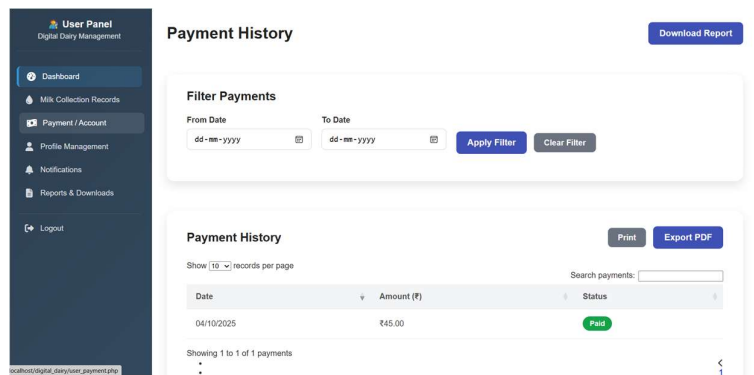
farmer pages



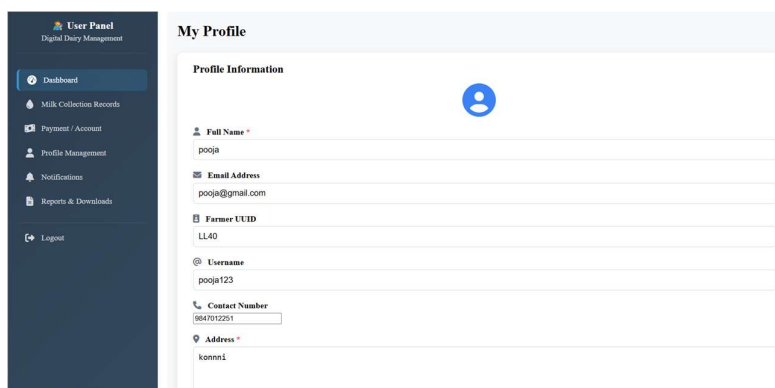
Farmer view- Personal farmer dashboard showing individual performance metrics



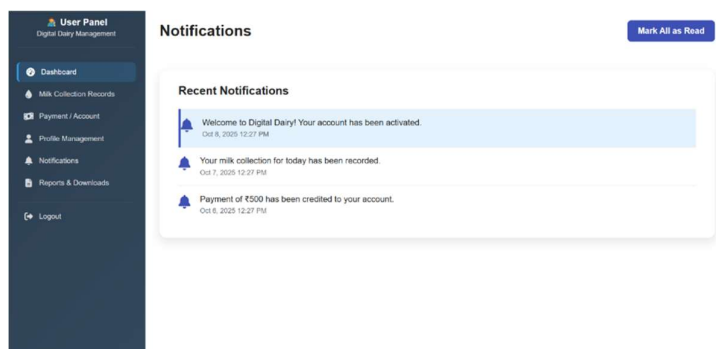
Framer view-dashboard showing milk record



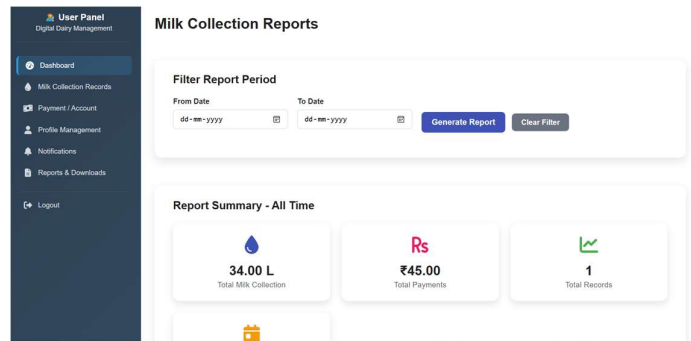
Farmer view- Payment history and earnings tracking for farmers



Farmer view- Farmer profile management interface



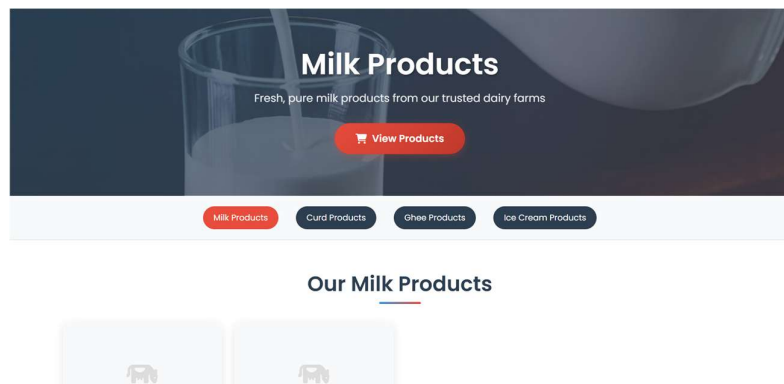
framer view- Notification center for farmers.



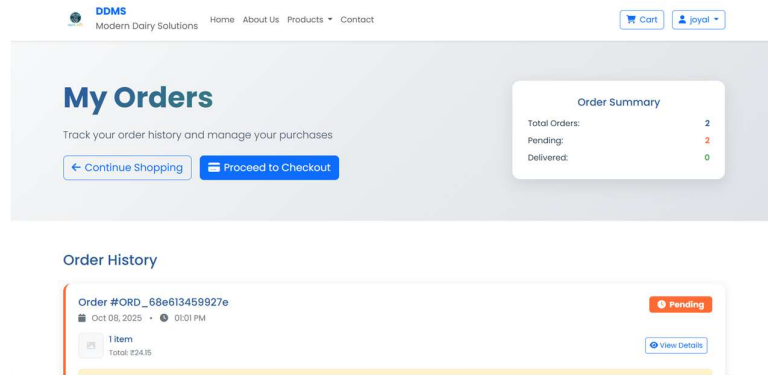
Farmer view- Personal reporting dashboard for farmers

Customer pages:

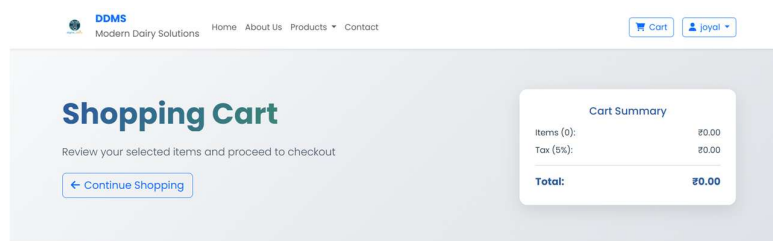
Customer view- Customer registration and account creation



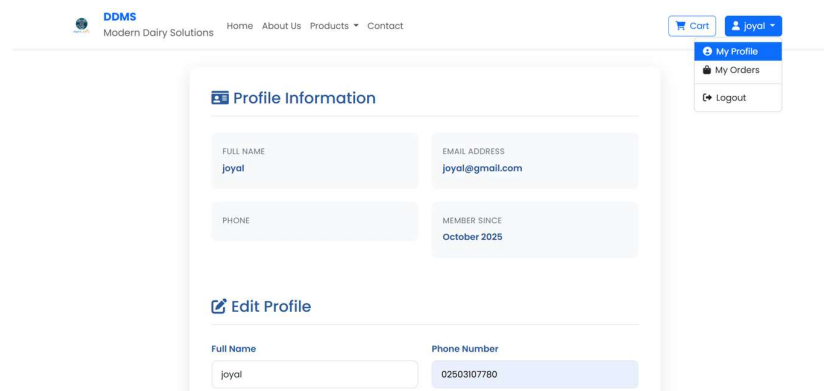
Customer view- Milk products catalog page



Customer view- Individual order detail view



Customer view- Shopping cart management interface



Customer view- Customer profile management page.

9.2 CONCLUSION

The Digital Dairy Management System (DDMS) successfully demonstrates how digital technology can transform traditional dairy operations into an efficient, automated, and transparent process. Developed using PHP, MySQL, HTML, CSS, and Bootstrap, the system provides a centralized platform that effectively manages **farmer details, milk collection, product inventory, sales records, and delivery management**. The project has met its objectives by simplifying data handling, reducing human errors, and improving the overall productivity of dairy operations. With the inclusion of real-time data processing and reporting features, the system ensures better decision-making for administrators and greater convenience for farmers and customers. Through this project, it was learned how to design, develop, and implement a full-stack web application using modern web technologies. The system, though efficient in its current scope, offers vast potential for future improvements such as mobile app integration, online payments, and advanced analytics. In conclusion, the **Digital Dairy Management System** is a reliable, user-friendly, and scalable solution that can serve as a strong foundation for the **digital transformation of the dairy sector**, particularly in regions like Kerala where cooperative and private dairies play a key role in local economies.

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2. Web Resources

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- <https://www.mysql.com/> – MySQL Database Reference
- <https://getbootstrap.com/> – Bootstrap UI Framework
- <https://fontawesome.com/> – Icon Resources
- <https://www.chartjs.org/> – Chart.js Library for Data Visualization
- https://github.com/Joyalsir/digital_dairy – Project Source Repository

3. Online Tutorials and References

- W3Schools – <https://www.w3schools.com/>
- Stack Overflow – <https://stackoverflow.com/>
- TutorialsPoint – <https://www.tutorialspoint.com/>